1.2 MATERIALS SCIENCE AND ENGINEERING

Sometimes it is useful to subdivide the discipline of materials science and engineering into *materials science* and *materials engineering* subdisciplines. Strictly speaking, materials science involves investigating the relationships that exist between the structures and properties of materials (i.e., why materials have their properties). In contrast, materials engineering involves, on the basis of these structure–property correlations, designing or engineering the structure of a material to produce a predetermined set of properties. From a functional perspective, the role of a materials scientist is to develop or synthesize new materials, whereas a materials engineer is called upon to create new products or systems using existing materials and/or to develop techniques for processing materials. Most graduates in materials programs are trained to be both materials scientists and materials engineers.

Structure is, at this point, a nebulous term that deserves some explanation. In brief, the structure of a material usually relates to the arrangement of its internal components. Structural elements may be classified on the basis of size and in this regard there are several levels:

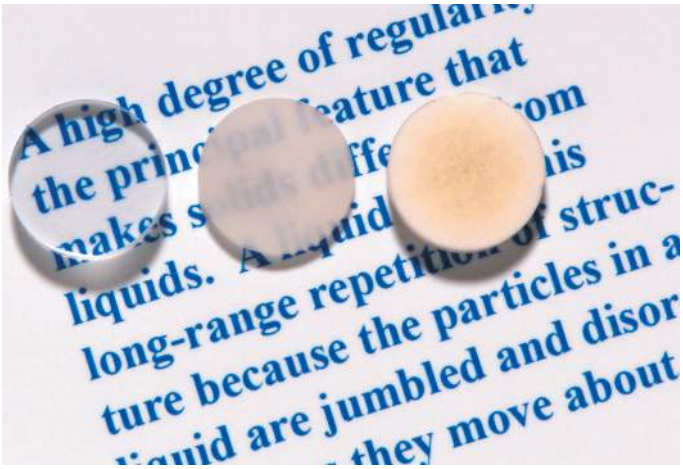
- Subatomic structure—involves electrons within the individual atoms, their energies and interactions with the nuclei.
- Atomic structure—relates to the organization of atoms to yield molecules or crystals.
- Nanostructure—deals with aggregates of atoms that form particles (nanoparticles) that have nanoscale dimensions (less that about 100 nm).
- Microstructure—those structural elements that are subject to direct observation using some type of microscope (structural features having dimensions between 100 nm and several millimeters).
- Macrostructure—structural elements that may be viewed with the naked eye (with scale range between several millimeters and on the order of a meter).

Atomic structure, nanostructure, and microstructure of materials are investigated using microscopic techniques discussed in Section 4.10.

The notion of *property* deserves elaboration. While in service use, all materials are exposed to external stimuli that evoke some types of responses. For example, a specimen subjected to forces experiences deformation, or a polished metal surface reflects light. A property is a material trait in terms of the kind and magnitude of response to a specific imposed stimulus. Generally, definitions of properties are made independent of material shape and size.

Virtually all important properties of solid materials may be grouped into six different categories: mechanical, electrical, thermal, magnetic, optical, and deteriorative. For each, there is a characteristic type of stimulus capable of provoking different responses. These are noted as follows:

- Mechanical properties—relate deformation to an applied load or force; examples include elastic modulus (stiffness), strength, and resistance to fracture.
- Electrical properties—the stimulus is an applied electric field; typical properties include electrical conductivity and dielectric constant.
- Thermal properties—are related to changes in temperature or temperature gradients across a material; examples of thermal behavior include thermal expansion and heat capacity.
- Magnetic properties—the responses of a material to the application of a magnetic field; common magnetic properties include magnetic susceptibility and magnetization.
- Optical properties—the stimulus is electromagnetic or light radiation; index of refraction and reflectivity are representative optical properties.



William D. Callister Jr./ Specimen preparation, P.A. Lessing

Figure 1.1 Three thin disk specimens of aluminum oxide that have been placed over a printed page in order to demonstrate their differences in light-transmittance characteristics. The disk on the left is *transparent* (i.e., virtually all light that is reflected from the page passes through it), whereas the one in the center is *translucent* (meaning that some of this reflected light is transmitted through the disk). The disk on the right is *opaque*—that is, none of the light passes through it. These differences in optical properties are a consequence of differences in structure of these materials, which have resulted from the way the materials were processed.

- Deteriorative characteristics—relate to the chemical reactivity of materials; for example, corrosion resistance of metals.

The chapters that follow discuss properties that fall within each of these six classifications.

In addition to structure and properties, two other important components are involved in the science and engineering of materials—namely, *processing* and *performance*. With regard to the relationships of these four components, the structure of a material depends on how it is processed. Furthermore, a material's performance is a function of its properties.

We present an example of these processing-structure-properties-performance principles in Figure 1.1, a photograph showing three thin disk specimens placed over some printed matter. It is obvious that the optical properties (i.e., the light transmittance) of each of the three materials are different; the one on the left is transparent (i.e., virtually all of the reflected light from the printed page passes through it), whereas the disks in the center and on the right are, respectively, translucent and opaque. All of these specimens are of the same material, aluminum oxide, but the leftmost one is what we call a *single crystal*—that is, has a high degree of perfection—which gives rise to its transparency. The center one is composed of numerous and very small single crystals that are all connected; the boundaries between these small crystals scatter a portion of the light reflected from the printed page, which makes this material optically translucent. Finally, the specimen on the right is composed not only of many small, interconnected crystals, but also of a large number of very small pores or void spaces. These pores scatter the reflected light to a greater degree than the crystal boundaries and render this material opaque. Thus, the structures of these three specimens are different in terms of crystal boundaries and pores, which affect the optical transmittance properties. Furthermore, each material was produced using a different processing technique. If optical transmittance is an important parameter relative to the ultimate in-service application, the performance of each material will be different.

This interrelationship among processing, structure, properties, and performance of materials may be depicted in linear fashion as in the schematic illustration shown in Figure 1.2. The model represented by this diagram has been called by some the *central paradigm of materials science and engineering* or sometimes just the *materials paradigm*. (The term "paradigm" means a model or set of ideas.) This paradigm, formulated in the 1990s is, in essence, the core of the discipline of materials science and engineering. It describes the protocol for selecting and designing materials for specific and well-defined



Figure 1.2 The four components of the discipline of materials science and engineering and their interrelationship.

applications, and has had a profound influence on the field of materials.[2] Previous to this time the materials science/engineering approach was to design components and systems using the existing palette of materials. The significance of this new paradigm is reflected in the following quotation: ". . . whenever a material is being created, developed, or produced, the properties or phenomena the material exhibits are of central concern. Experience shows that the properties and phenomena associated with a material are intimately related to its composition and structure at all levels, including which atoms are present and how the atoms are arranged in the material, and that this structure is the result of synthesis and processing."[3]

Throughout this text, we draw attention to the relationships among these four components in terms of the design, production, and utilization of materials.

1.3 WHY STUDY MATERIALS SCIENCE AND ENGINEERING?

Why do engineers and scientists study materials? Simply, because things engineers design are made of materials. Many an applied scientist or engineer (e.g., mechanical, civil, chemical, electrical), is at one time or another exposed to a design problem involving materials—for example, a transmission gear, the superstructure for a building, an oil refinery component, or an integrated circuit chip. Of course, materials scientists and engineers are specialists who are totally involved in the investigation and design of materials.

Many times, an engineer has the option of selecting a best material from the thousands available. The final decision is normally based on several criteria. First, the in-service conditions must be characterized, for these dictate the properties required of the material. Only on rare occasions does a material possess the optimum or ideal combination of properties. Thus, it may be necessary to trade one characteristic for another. The classic example involves strength and ductility; normally, a material having a high strength has only a limited ductility. In such cases, a reasonable compromise between two or more properties may be necessary.

A second selection consideration is any deterioration of material properties that may occur during service operation. For example, significant reductions in mechanical strength may result from exposure to elevated temperatures or corrosive environments.

Finally, probably the overriding consideration is that of economics: What will the finished product cost? A material may be found that has the optimum set of properties but is prohibitively expensive. Here again, some compromise is inevitable. The cost of a finished piece also includes any expense incurred during fabrication to produce the desired shape.

The more familiar an engineer or scientist is with the various characteristics and structure–property relationships, as well as the processing techniques of materials, the more proficient and confident he or she will be in making judicious materials choices based on these criteria.

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[2]This paradigm has recently been updated to include the component of material sustainability in the "Modified Paradigm of Materials Science and Engineering," as represented by the following diagram:

Processing → Structure → Properties → Performance → Reuse/Recyclability

[3]"*Materials Science and Engineering for the 1990s*," p. 27, National Academies Press, Washington, DC, 1998.

CASE STUDY 1.1

Liberty Ship Failures

The following case study illustrates one role that materials scientists and engineers are called upon to assume in the area of materials performance: analyze mechanical failures, determine their causes, and then propose appropriate measures to guard against future incidents.

The failure of many of the World War II Liberty ships[4] is a well-known and dramatic example of the brittle fracture of steel that was thought to be ductile.[5] Some of the early ships experienced structural damage when cracks developed in their decks and hulls. Three of them catastrophically split in half when cracks formed, grew to critical lengths, and then rapidly propagated completely around the ships' girths. Figure 1.3 shows one of the ships that fractured the day after it was launched.

Subsequent investigations concluded one or more of the following factors contributed to each failure:[6]

- When some normally ductile metal alloys are cooled to relatively low temperatures, they become susceptible to brittle fracture—that is, they experience a ductile-to-brittle transition upon cooling through a critical range of temperatures.

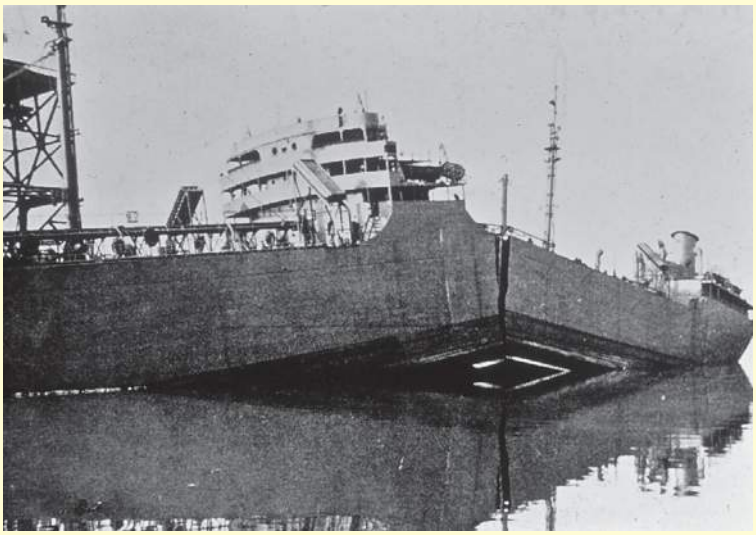


Figure 1.3 The Liberty ship *S.S. Schenectady,* which, in 1943, failed before leaving the shipyard.
(Reprinted with permission of Earl R. Parker, *Brittle Behavior of Engineering Structures,* National Academy of Sciences, National Research Council, John Wiley & Sons, New York, 1957.)

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[4]During World War II, 2,710 Liberty cargo ships were mass-produced by the United States to supply food and materials to the combatants in Europe.

[5]Ductile metals fail after relatively large degrees of permanent deformation; however, very little if any permanent deformation accompanies the fracture of brittle materials. Brittle fractures can occur very suddenly as cracks spread rapidly; crack propagation is normally much slower in ductile materials, and the eventual fracture takes longer. For these reasons, the ductile mode of fracture is usually preferred. Ductile and brittle fractures are discussed in Sections 8.3 and 8.4.

[6]Sections 8.2 through 8.6 discuss various aspects of failure.

These Liberty ships were constructed of steel that experienced a ductile-to-brittle transition. Some of them were deployed to the frigid North Atlantic, where the once ductile metal experienced brittle fracture when temperatures dropped to below the transition temperature.[7]

- The corner of each hatch (i.e., door) was square; these corners acted as points of stress concentration where cracks can form.
- German U-boats were sinking cargo ships faster than they could be replaced using existing construction techniques. Consequently, it became necessary to revolutionize construction methods to build cargo ships faster and in greater numbers. This was accomplished using prefabricated steel sheets that were assembled by welding rather than by the traditional time-consuming riveting. Unfortunately, cracks in welded structures may propagate unimpeded for large distances, which can lead to catastrophic failure. However, when structures are riveted, a crack ceases to propagate once it reaches the edge of a steel sheet.
- Weld defects and *discontinuities* (i.e., sites where cracks can form) were introduced by inexperienced operators.

Remedial measures taken to correct these problems included the following:

- Lowering the ductile-to-brittle temperature of the steel to an acceptable level by improving steel quality (e.g., reducing sulfur and phosphorus impurity contents).
- Rounding off hatch corners by welding a curved reinforcement strip on each corner.[8]
- Installing crack-arresting devices such as riveted straps and strong weld seams to stop propagating cracks.
- Improving welding practices and establishing welding codes.

In spite of these failures, the Liberty ship program was considered a success for several reasons, the primary reason being that ships that survived failure were able to supply Allied Forces in the theater of operations and in all likelihood shortened the war. In addition, structural steels were developed with vastly improved resistances to catastrophic brittle fractures. Detailed analyses of these failures advanced the understanding of crack formation and growth, which ultimately evolved into the discipline of fracture mechanics.

[7]This ductile-to-brittle transition phenomenon, as well as techniques that are used to measure and raise the critical temperature range, are discussed in Section 8.6.

[8]The reader may note that corners of windows and doors for all of today's marine and aircraft structures are rounded.

1.4 CLASSIFICATION OF MATERIALS

Solid materials have been conveniently grouped into three basic categories: metals, ceramics, and polymers, a scheme based primarily on chemical makeup and atomic structure. Most materials fall into one distinct grouping or another. In addition, there are the composites that are engineered combinations of two or more different materials. A brief explanation of these material classifications and representative characteristics is offered next. Another category is advanced materials—those used in high-technology applications, such as semiconductors, biomaterials, smart materials, and nanoengineered materials; these are discussed in Section 1.5.

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Tutorial Video: What Are the Different Classes of Materials?

Metals

Metals are composed of one or more metallic elements (e.g., iron, aluminum, copper, titanium, gold, nickel), and often also nonmetallic elements (e.g., carbon, nitrogen, oxygen) in relatively small amounts.[9] Atoms in metals and their alloys are arranged in a

[9]The term *metal alloy* refers to a metallic substance that is composed of two or more elements.

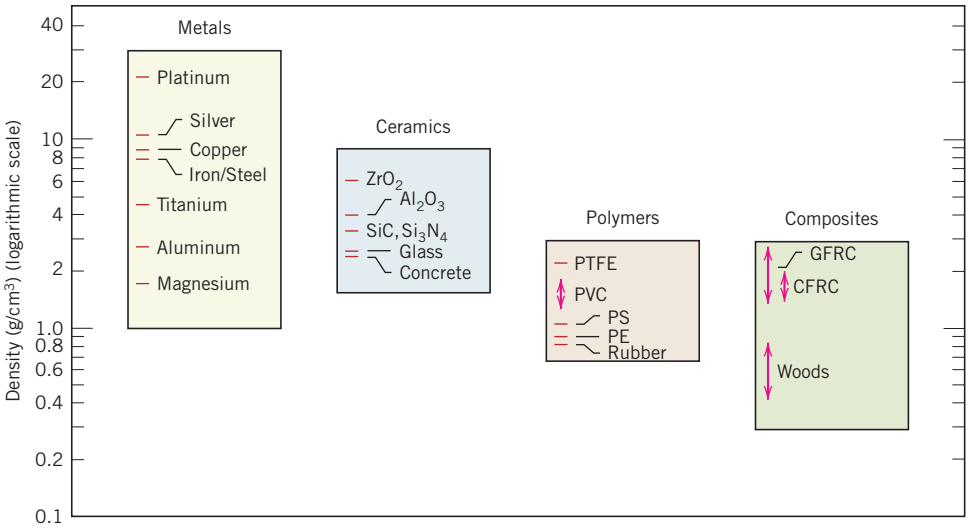


Figure 1.4 Bar chart of room-temperature density values for various metals, ceramics, polymers, and composite materials.

very orderly manner (as discussed in Chapter 3) and are relatively dense in comparison to the ceramics and polymers (Figure 1.4). With regard to mechanical characteristics, these materials are relatively stiff (Figure 1.5) and strong (Figure 1.6), yet are ductile (i.e., capable of large amounts of deformation without fracture), and are resistant to fracture (Figure 1.7), which accounts for their widespread use in structural applications. Metallic materials have large numbers of nonlocalized electrons—that is, these electrons are not bound to particular atoms. Many properties of metals are directly attributable to these electrons. For example, metals are extremely good conductors of electricity (Figure 1.8) and heat, and are not transparent to visible light; a polished metal surface has a lustrous appearance. In addition, some of the metals (i.e., Fe, Co, and Ni) have desirable magnetic properties.

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Tutorial Video: Metals

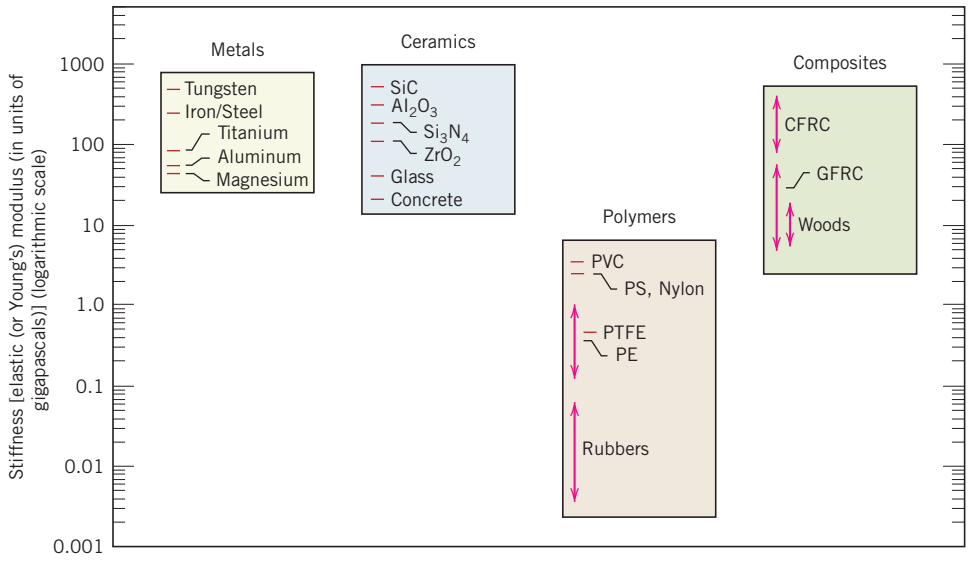


Figure 1.5 Bar chart of room-temperature stiffness (i.e., elastic modulus) values for various metals, ceramics, polymers, and composite materials.

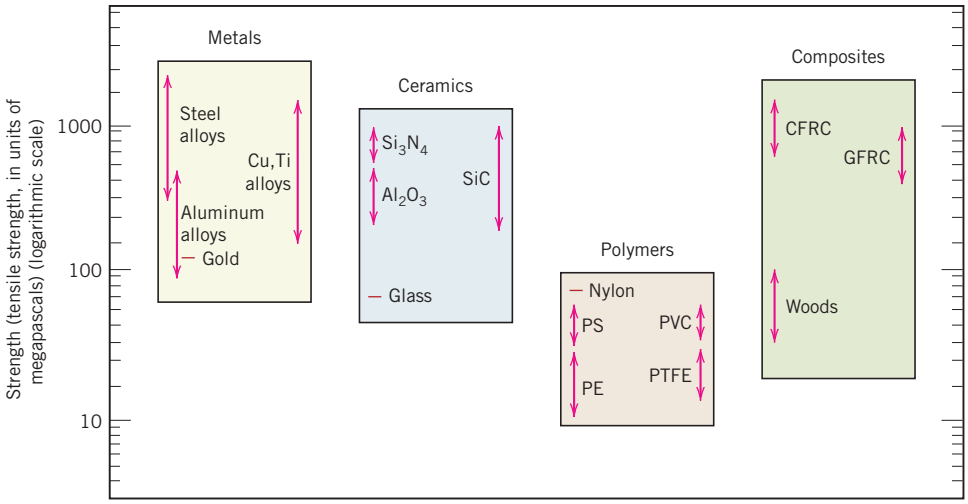


Figure 1.6 Bar chart of room-temperature strength (i.e., tensile strength) values for various metals, ceramics, polymers, and composite materials.

Figure 1.9 shows several common and familiar objects that are made of metallic materials. Furthermore, the types and applications of metals and their alloys are discussed in Chapter 11.

Ceramics

Ceramics are compounds between metallic and nonmetallic elements; they are most frequently oxides, nitrides, and carbides. For example, common ceramic materials include aluminum oxide (or *alumina,* Al_2O_3), silicon dioxide (or *silica,* SiO_2), silicon carbide (SiC), silicon nitride (Si_3N_4), and, in addition, what some refer to as the *traditional ceramics*—those composed of clay minerals (e.g., porcelain), as well as cement and glass. With regard to mechanical behavior, ceramic materials are relatively stiff and strong—stiffnesses and strengths are comparable to those of the metals (Figures 1.5 and 1.6). In addition, they are typically very hard. Historically, ceramics have exhibited extreme brittleness (lack of ductility) and are highly susceptible to fracture (Figure 1.7). However, newer ceramics are being engineered to have improved resistance to fracture; these materials are used for cookware, cutlery, and even automobile engine parts. Furthermore, ceramic materials are typically insulative to the

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Tutorial Video: Ceramics

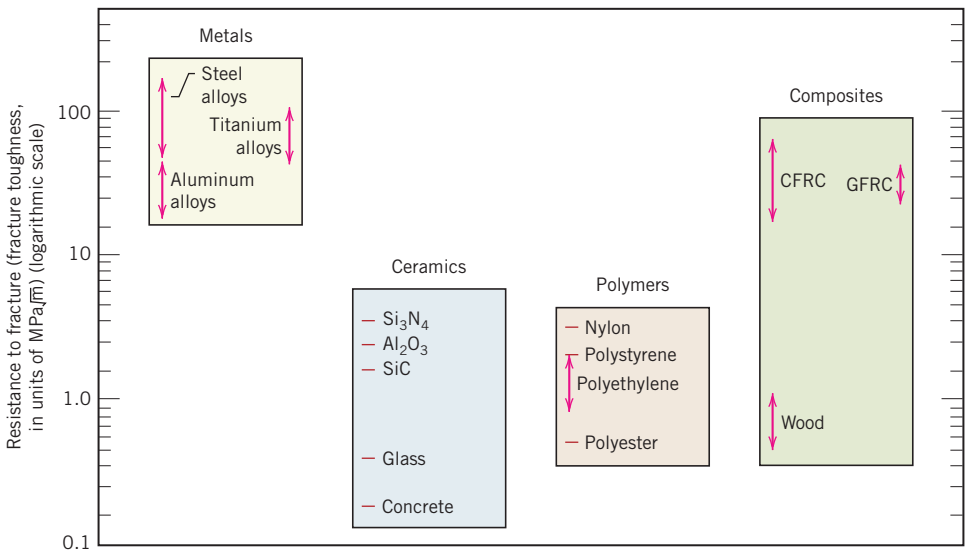


Figure 1.7 Bar chart of room-temperature resistance to fracture (i.e., fracture toughness) for various metals, ceramics, polymers, and composite materials. (Reprinted from *Engineering Materials 1: An Introduction to Properties, Applications and Design,* third edition, M. F. Ashby and D. R. H. Jones, pages 177 and 178. Copyright 2005, with permission from Elsevier.)

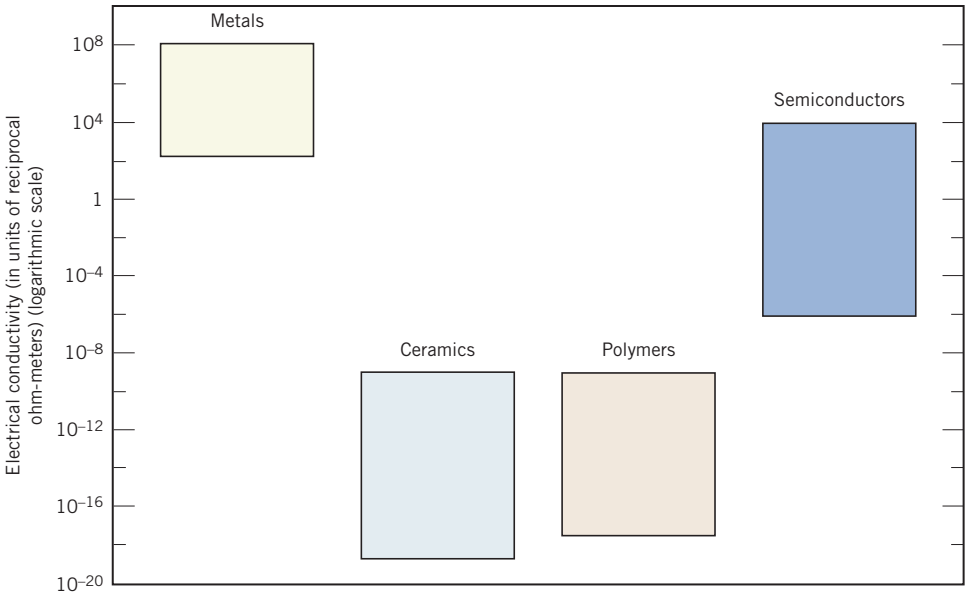


Figure 1.8 Bar chart of room-temperature electrical conductivity ranges for metals, ceramics, polymers, and semiconducting materials.

passage of heat and electricity (i.e., have low electrical conductivities, Figure 1.8) and are more resistant to high temperatures and harsh environments than are metals and polymers. With regard to optical characteristics, ceramics may be transparent, translucent, or opaque (Figure 1.1), and some of the oxide ceramics (e.g., Fe_3O_4) exhibit magnetic behavior.

Several common ceramic objects are shown in Figure 1.10. The characteristics, types, and applications of this class of materials are also discussed in Chapters 12 and 13.

Polymers

Polymers include the familiar plastic and rubber materials. Many of them are organic compounds that are chemically based on carbon, hydrogen, and other nonmetallic elements (i.e., O, N, and Si). Furthermore, they have very large molecular structures, often chainlike in nature, that often have a backbone of carbon atoms. Some common and familiar polymers are polyethylene (PE), nylon, poly(vinyl chloride) (PVC), polycarbonate (PC), polystyrene (PS), and silicone rubber. These materials typically have low densities (Figure 1.4), whereas their mechanical characteristics are generally dissimilar to those of the metallic and ceramic materials—they are not as stiff or strong as these



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Figure 1.9 Familiar objects made of metals and metal alloys (from left to right): silverware (fork and knife), scissors, coins, a gear, a wedding ring, and a nut and bolt.



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Figure 1.10 Common objects made of ceramic materials: scissors, a china teacup, a building brick, a floor tile, and a glass vase.

other material types (Figures 1.5 and 1.6). However, on the basis of their low densities, many times their stiffnesses and strengths on a per-mass basis are comparable to those of the metals and ceramics. In addition, many of the polymers are extremely ductile and pliable (i.e., plastic), which means they are easily formed into complex shapes. In general, they are relatively inert chemically and unreactive in a large number of environments. Furthermore, they have low electrical conductivities (Figure 1.8) and are nonmagnetic. One major drawback to the polymers is their tendency to soften and/or decompose at modest temperatures, which, in some instances, limits their use.

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Tutorial Video: Polymers

Figure 1.11 shows several articles made of polymers that are familiar to the reader. Chapters 14 and 15 are devoted to discussions of the structures, properties, applications, and processing of polymeric materials.



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Figure 1.11 Several common objects made of polymeric materials: plastic tableware (spoon, fork, and knife), billiard balls, a bicycle helmet, two dice, a lawn mower wheel (plastic hub and rubber tire), and a plastic milk carton.

CASE STUDY 1.2

Carbonated Beverage Containers

One common item that presents some interesting material property requirements is the container for carbonated beverages. The material used for this application must satisfy the following constraints: (1) provide a barrier to the passage of carbon dioxide, which is under pressure in the container; (2) be nontoxic, unreactive with the beverage, and, preferably, recyclable; (3) be relatively strong and capable of surviving a drop from a height of several feet when containing the beverage; (4) be inexpensive, including the cost to fabricate the final shape; (5) if optically transparent, retain its optical clarity; and (6) be capable of being produced in different colors and/or adorned with decorative labels.

All three of the basic material types—metal (aluminum), ceramic (glass), and polymer (polyester plastic)—are used for carbonated beverage containers (per the chapter-opening photographs). All of these materials are nontoxic and unreactive with beverages. In addition, each material has its pros and cons. For example, the aluminum alloy is relatively strong (but easily dented), is a very good barrier to the diffusion of carbon dioxide, is easily recycled, cools beverages rapidly, and allows labels to be painted onto its surface. However, the cans are optically opaque and relatively expensive to produce. Glass is impervious to the passage of carbon dioxide, is a relatively inexpensive material, and may be recycled, but it cracks and fractures easily, and glass bottles are relatively heavy. Whereas plastic is relatively strong, may be made optically transparent, is inexpensive and lightweight, and is recyclable, it is not as impervious to the passage of carbon dioxide as aluminum and glass. For example, you may have noticed that beverages in aluminum and glass containers retain their carbonization (i.e., "fizz") for several years, whereas those in two-liter plastic bottles "go flat" within a few months.

Composites

A *composite* is composed of two (or more) individual materials that come from the categories previously discussed—metals, ceramics, and polymers. The design goal of a composite is to achieve a combination of properties that is not displayed by any single material and also to incorporate the best characteristics of each of the component materials. A large number of composite types are represented by different combinations of metals, ceramics, and polymers. Furthermore, some naturally occurring materials are composites—for example, wood and bone. However, most of those we consider in our discussions are synthetic (or human-made) composites.

One of the most common and familiar composites is fiberglass, in which small glass fibers are embedded within a polymeric material (normally an epoxy or polyester).[10] The glass fibers are relatively strong and stiff (but also brittle), whereas the polymer is more flexible. Thus, fiberglass is relatively stiff, strong (Figures 1.5 and 1.6), and flexible. In addition, it has a low density (Figure 1.4).

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Tutorial Video: Composites

Another technologically important material is the carbon fiber–reinforced polymer (CFRP) composite—carbon fibers that are embedded within a polymer. These materials are stiffer and stronger than glass fiber–reinforced materials (Figures 1.5 and 1.6) but more expensive. CFRP composites are used in some aircraft and aerospace applications, as well as in high-tech sporting equipment (e.g., bicycles, golf clubs, tennis rackets, skis/snowboards) and recently in automobile bumpers. The new Boeing 787 fuselage is primarily made from such CFRP composites.

Chapter 16 is devoted to a discussion of these interesting composite materials.

[10]Fiberglass is sometimes also termed a *glass fiber–reinforced polymer* composite (GFRP).

There is an alternative and more illustrative way of presenting property values by material type than was portrayed by Figures 1.4 through 1.8—that is, if we plot the values of one property versus those of another property for a large number of different types of materials. Both axes are scaled logarithmically and usually span several (at least three) orders of magnitude, so as to include the properties of virtually all materials. For example, Figure 1.12 is one such diagram; here logarithm of stiffness (modulus of elasticity or Young's modulus) is plotted versus the logarithm of density. Here it may be noted that data values for a specific type (or "family") of materials (e.g., metals, ceramics, polymers) cluster together and are enclosed within an envelope (or "bubble") delineated with a bold line; hence, each of these envelopes defines the property range for its material family.

This is a simple, comprehensive, and concise display of the kind of information contained in both Figures 1.4 and 1.5 that shows how density and stiffness correlate with one another among the various kinds of materials. Charts such as Figure 1.12 may be constructed for any two material properties—for example, thermal conductivity versus electrical conductivity. Thus, a relatively large number of plots of this type are available given the possible combinations of pairs of the various material properties. They are

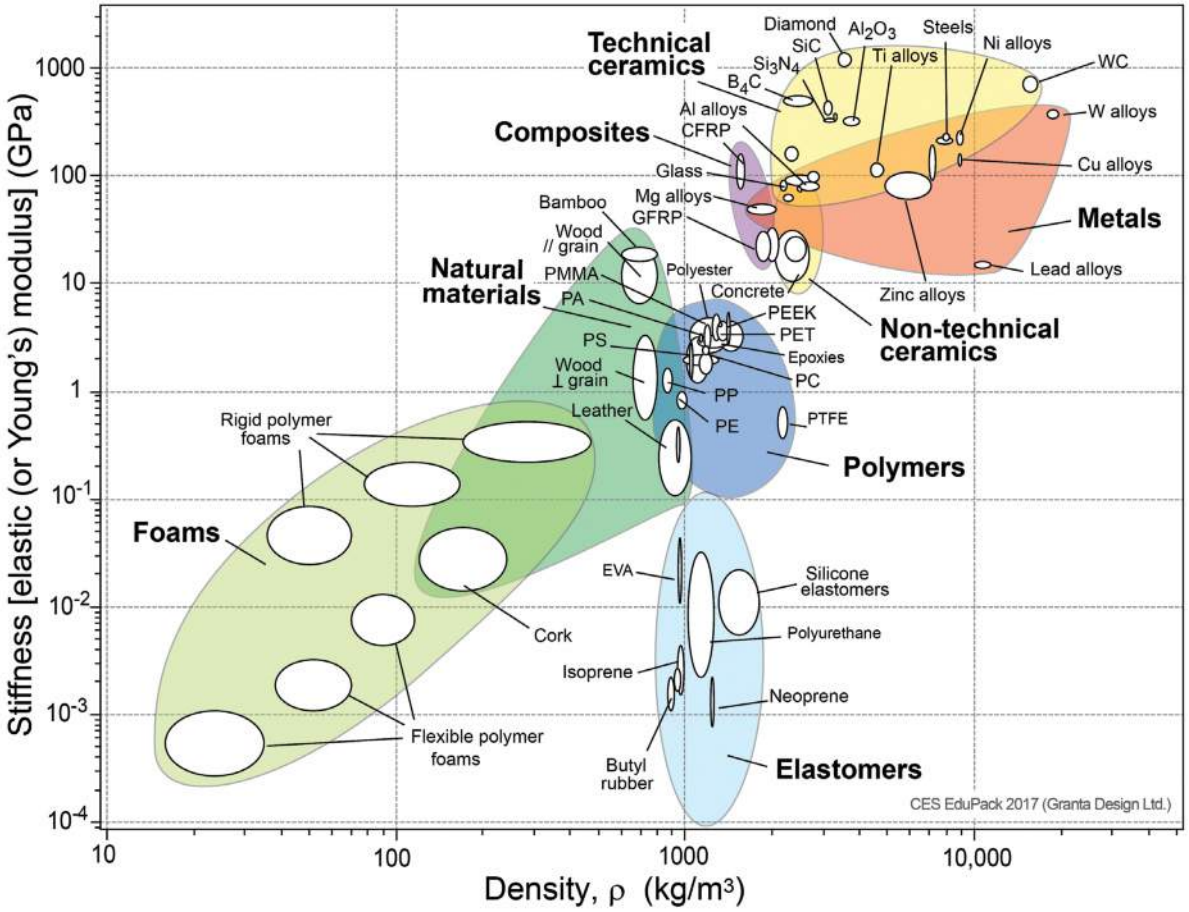


Figure 1.12 Modulus of elasticity (stiffness) versus density materials selection chart.
(Chart created using CES EduPack 2017, Granta Design Ltd.)

often referred to as "materials property charts," "materials selection charts," "bubble charts," or "Ashby charts" (after Michael F. Ashby, who developed them).[11]

In Figure 1.12, envelopes for three important engineering material families are included that were not discussed previously in this section. These are as follows:

- Elastomers—polymeric materials that display rubbery-like behavior (high degrees of elastic deformation).
- Natural materials—those that occur in nature; for example, wood, leather, and cork.
- Foams—typically polymeric materials that have high porosities (contain a large volume fraction of small pores), which are often used for cushions and packaging.

These bubble charts are extremely useful tools in engineering design and are used extensively in the materials selection process in both academia and industry.[12] When considering materials for products, an engineer is often confronted with competing objectives (e.g., light weight and stiffness) and must be in a position to assess possible trade-offs among any competing requirements. Insights into the consequences of trade-off choices may be gleaned by using appropriate bubble charts. This procedure is demonstrated in the *Materials Selection for a Torsionally Stressed Cylindrical Shaft* case study found in both the "Library of Case Studies" [which may be found in WileyPLUS or at *www.wiley.com/college/callister* (Student Companion Site)] and the "Mechanical Engineering Online Support Module" [which may be found in all digital version of the text or at *www.wiley.com/college/callister (Student Companion Site)*].

1.5 ADVANCED MATERIALS

Materials utilized in high-technology (or high-tech) applications are sometimes termed *advanced materials.* By *high technology,* we mean a device or product that operates or functions using relatively intricate and sophisticated principles, including electronic equipment (cell phones, DVD players, etc.), computers, fiber-optic systems, high-energy density batteries, energy-conversion systems, and aircraft. These advanced materials are typically traditional materials whose properties have been enhanced and also newly developed, high-performance materials. Furthermore, they may be of all material types (e.g., metals, ceramics, polymers) and are normally expensive. Advanced materials include semiconductors, biomaterials, and what we may term *materials of the future* (i.e., smart materials and nanoengineered materials), which we discuss next. The properties and applications of a number of these advanced materials—for example, materials that are used for lasers, batteries, magnetic information storage, liquid crystal displays (LCDs), and fiber optics—are also discussed in subsequent chapters.

Semiconductors

Semiconductors have electrical properties that are intermediate between those of electrical conductors (i.e., metals and metal alloys) and insulators (i.e., ceramics and polymers)—see Figure 1.8. Furthermore, the electrical characteristics of these materials are extremely sensitive to the presence of minute concentrations of impurity atoms, for which the concentrations may be controlled over very small spatial regions. Semiconductors have made possible the advent of integrated circuitry that has totally revolutionized the electronics and computer industries (not to mention our lives) over the past four decades.

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[11]A collection of these charts may be found at the following web address: www.teachingresources.grantadesign.com/charts.

[12]Granta Design's CES EduPack is an excellent software package for teaching the principles of materials selection in design using these bubble charts.

Biomaterials

The length and the quality of our lives are being extended and improved, in part, due to advancements in the ability to replace diseased and injured body parts. Replacement implants are constructed of *biomaterials*—nonviable (i.e., nonliving) materials that are implanted into the body, so that they function in a reliable, safe, and physiologically satisfactory manner, while interacting with living tissue. That is, biomaterials must be *biocompatible*—compatible with body tissues and fluids with which they are in contact over acceptable time periods. Biocompatible materials must neither elicit rejection or physiologically unacceptable responses nor release toxic substances. Consequently, some rather stringent constraints are imposed on materials in order for them to be biocompatible.

Suitable biomaterials are to be found among the several classes of materials discussed earlier in this chapter—i.e., metal alloys, ceramics, polymers, and composite materials. Throughout the remainder of this book we draw the reader's attention to those materials that are used in biotechnology applications.

Over the past several years the development of new and better biomaterials has accelerated rapidly; today, this is one of the "hot" materials areas, with an abundance of new, exciting, and high-salary job opportunities. Example biomaterial applications include joint (e.g., hip, knee) and heart valve replacements, vascular (blood vessel) grafts, fracture-fixation devices, dental restorations, and generation of new organ tissues.

Smart Materials

Smart (or *intelligent*) *materials* are a group of new and state-of-the-art materials now being developed that will have a significant influence on many of our technologies. The adjective *smart* implies that these materials are able to sense changes in their environment and then respond to these changes in predetermined manners—traits that are also found in living organisms. In addition, this *smart* concept is being extended to rather sophisticated systems that consist of both smart and traditional materials.

Components of a smart material (or system) include some type of sensor (which detects an input signal) and an actuator (which performs a responsive and adaptive function). Actuators may be called upon to change shape, position, natural frequency, or mechanical characteristics in response to changes in temperature, electric fields, and/or magnetic fields.

Four types of materials are commonly used for actuators: shape-memory alloys, piezoelectric ceramics, magnetostrictive materials, and electrorheological/magnetorheological fluids. *Shape-memory alloys* are metals that, after having been deformed, revert to their original shape when temperature is changed (see the Materials of Importance box following Section 10.9). *Piezoelectric ceramics* expand and contract in response to an applied electric field (or voltage); conversely, they also generate an electric field when their dimensions are altered (see Section 18.25). The behavior of *magnetostrictive materials* is analogous to that of the piezoelectrics, except that they are responsive to magnetic fields. Also, *electrorheological* and *magnetorheological fluids* are liquids that experience dramatic changes in viscosity upon the application of electric and magnetic fields, respectively.

Materials/devices employed as sensors include optical fibers (Section 21.14), piezoelectric materials (including some polymers), and microelectromechanical systems (MEMS; Section 13.10).

For example, one type of smart system is used in helicopters to reduce aerodynamic cockpit noise created by the rotating rotor blades. Piezoelectric sensors inserted into the blades monitor blade stresses and deformations; feedback signals from these sensors are fed into a computer-controlled adaptive device that generates noise-canceling antinoise.

Nanomaterials

One new material class that has fascinating properties and tremendous technological promise is the *nanomaterials,* which may be any one of the four basic types—metals,

ceramics, polymers, or composites. However, unlike these other materials, they are not distinguished on the basis of their chemistry but rather their size; the *nano* prefix denotes that the dimensions of these structural entities are on the order of a nanometer (10^{-9} m)—as a rule, less than 100 nanometers (nm; equivalent to the diameter of approximately 500 atoms).

Prior to the advent of nanomaterials, the general procedure scientists used to understand the chemistry and physics of materials was to begin by studying large and complex structures and then investigate the fundamental building blocks of these structures that are smaller and simpler. This approach is sometimes termed *top-down science*. However, with the development of scanning probe microscopes (Section 4.10), which permit observation of individual atoms and molecules, it has become possible to design and build new structures from their atomic-level constituents, one atom or molecule at a time (i.e., "materials by design"). This ability to arrange atoms carefully provides opportunities to develop mechanical, electrical, magnetic, and other properties that are not otherwise possible. We call this the *bottom-up approach*, and the study of the properties of these materials is termed *nanotechnology*.[13]

Some of the physical and chemical characteristics exhibited by matter may experience dramatic changes as particle size approaches atomic dimensions. For example, materials that are opaque in the macroscopic domain may become transparent on the nanoscale; some solids become liquids, chemically stable materials become combustible, and electrical insulators become conductors. Furthermore, properties may depend on size in this nanoscale domain. Some of these effects are quantum mechanical in origin, whereas others are related to *surface phenomena*—the proportion of atoms located on surface sites of a particle increases dramatically as its size decreases.

Because of these unique and unusual properties, nanomaterials are finding niches in electronic, biomedical, sporting, energy production, and other industrial applications. Some are discussed in this text, including the following:

- Catalytic converters for automobiles (Materials of Importance box, Chapter 4)
- Nanocarbons—fullerenes, carbon nanotubes, and graphene (Section 13.10)
- Particles of carbon black as reinforcement for automobile tires (Section 16.2)
- Nanocomposites (Section 16.16)
- Magnetic nanosize grains that are used for hard disk drives (Section 20.11)
- Magnetic particles that store data on magnetic tapes (Section 20.11)

Whenever a new material is developed, its potential for harmful and toxicological interactions with humans and animals must be considered. Small nanoparticles have exceedingly large surface area–to–volume ratios, which can lead to high chemical reactivities. Although the safety of nanomaterials is relatively unexplored, there are concerns that they may be absorbed into the body through the skin, lungs, and digestive tract at relatively high rates, and that some, if present in sufficient concentrations, will pose health risks—such as damage to DNA or promotion of lung cancer.

1.6 MODERN MATERIALS' NEEDS

In spite of the tremendous progress that has been made in the discipline of materials science and engineering within the past few years, technological challenges remain, including the development of even more sophisticated and specialized materials, as well as consideration of the environmental impact of materials production. Some comment is appropriate relative to these issues so as to round out this perspective.

[13]One legendary and prophetic suggestion as to the possibility of nanoengineered materials was offered by Richard Feynman in his 1959 American Physical Society lecture titled "There's Plenty of Room at the Bottom."

A number of today's important technological sectors involve energy. There is a recognized need to find new and economical sources of energy, especially renewable energy, and to use present resources more efficiently. Materials will undoubtedly play a significant role in these developments—for example, the direct conversion of solar power into electrical energy. Solar cells employ some rather complex and expensive materials. To ensure a viable technology, materials that are highly efficient in this conversion process yet less costly must be developed.

In conjunction with improved solar cell materials, there is also a marked need for new materials for batteries that provide higher electrical energy-storage densities than those presently available and at lower costs. The current cutting-edge technology uses lithium ion batteries; these offer relatively high storage densities, but also present some technological challenges.

Significant quantities of energy are involved in transportation. Reducing the weight of transportation vehicles (automobiles, aircraft, trains, etc.), as well as increasing engine operating temperatures, will enhance fuel efficiency. New high-strength, low-density structural materials remain to be developed, as well as materials that have higher-temperature capabilities, for use in engine components.

The hydrogen fuel cell is another very attractive and feasible energy-conversion technology that has the advantage of being nonpolluting. It is just beginning to be implemented in batteries for electronic devices and holds promise as a power plant for automobiles. New materials still need to be developed for more efficient fuel cells and also for better catalysts to be used in the production of hydrogen.

Nuclear energy holds some promise, but the solutions to the many problems that remain necessarily involve materials, such as fuels, containment structures, and facilities for the disposal of radioactive waste.

Furthermore, environmental quality depends on our ability to control air and water pollution. Pollution control techniques employ various materials. In addition, materials processing and refinement methods need to be improved so that they produce less environmental degradation—that is, less pollution and less despoilage of the landscape from the mining of raw materials. Also, in some materials manufacturing processes, toxic substances are produced, and the ecological impact of their disposal must be considered.

Many materials that we use are derived from resources that are nonrenewable—that is, not capable of being regenerated, including most polymers, for which the prime raw material is oil, and some metals. These nonrenewable resources are gradually becoming depleted, which necessitates (1) the discovery of additional reserves, (2) the development of new materials having comparable properties with less adverse environmental impact, and/or (3) increased recycling efforts and the development of new recycling technologies. As a consequence of the economics of not only production but also environmental impact and ecological factors, it is becoming increasingly important to consider the "cradle-to-grave" life cycle of materials relative to the overall manufacturing process.

The roles that materials scientists and engineers play relative to these, as well as other environmental and societal issues, are discussed in more detail in Chapter 22.

SUMMARY

Materials Science and Engineering

- Six different property classifications of materials determine their applicability: mechanical, electrical, thermal, magnetic, optical, and deteriorative.
- One important relationship in the science of materials is the dependence of a material's properties on its structural elements. By *structure*, we mean how the internal component(s) of the material is (are) arranged. In terms of (and with increasing) dimensionality, structural elements include subatomic, atomic, nanoscopic, microscopic, and macroscopic.

- With regard to the design, production, and utilization of materials, there are four elements to consider—processing, structure, properties, and performance. The performance of a material depends on its properties, which in turn are a function of its structure(s); structure(s) is (are) determined by how the material was processed. The interrelationship among these four elements is sometimes called the central paradigm of materials science and engineering.
- Three important criteria in materials selection are in-service conditions to which the material will be subjected, any deterioration of material properties during operation, and economics or cost of the fabricated piece.

Classification of Materials

- On the basis of chemistry and atomic structure, materials are classified into three general categories: metals (metallic elements), ceramics (compounds between metallic and nonmetallic elements), and polymers (compounds composed of carbon, hydrogen, and other nonmetallic elements). In addition, composites are composed of at least two different material types.

Advanced Materials

- Another materials category is the advanced materials that are used in high-tech applications, including semiconductors (having electrical conductivities intermediate between those of conductors and insulators), biomaterials (which must be compatible with body tissues), smart materials (those that sense and respond to changes in their environments in predetermined manners), and nanomaterials (those that have structural features on the order of a nanometer, some of which may be designed on the atomic/molecular level).

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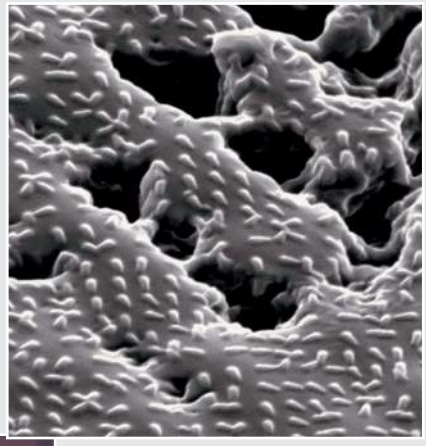
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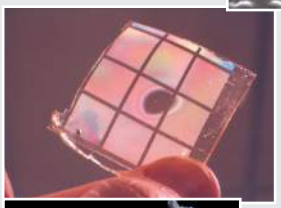
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Chapter 2 Atomic Structure and Interatomic Bonding



Courtesy Jeffrey Karp, Robert Langer and Alex Galakatos



Courtesy Jeffrey Karp, Robert Langer and Alex Galakatos



Paul D. Stewart/Science Source

The photograph at the bottom of this page is of a gecko.

Geckos, harmless tropical lizards, are extremely fascinating and extraordinary animals. They have very sticky feet (one of which is shown in the third photograph) that cling to virtually any surface. This characteristic makes it possible for them to run rapidly up vertical walls and along the undersides of horizontal surfaces. In fact, a gecko can support its body mass with a single toe! The secret to this remarkable ability is the presence of an extremely large number of microscopically small hairs on each of their toe pads. When these hairs come in contact with a surface, weak forces of attraction (i.e., van der Waals forces) are established between hair molecules and molecules on the surface. The fact that these hairs are so small and so numerous explains why the gecko grips surfaces so tightly. To release its grip, the gecko simply curls up its toes and peels the hairs away from the surface.

Using their knowledge of this mechanism of adhesion, scientists have developed several ultrastrong synthetic adhesives, one of which is an adhesive tape (shown in the second photograph) that is an especially promising tool for use in surgical procedures as a replacement for sutures and staples to close wounds and incisions. This material retains its adhesive nature in wet environments, is biodegradable, and does not release toxic substances as it dissolves during the healing process. Microscopic features of this adhesive tape are shown in the top photograph.



Barbara Peacock/Photodisc/Getty Images, Inc.

WHY STUDY *Atomic Structure and Interatomic Bonding?*

An important reason to have an understanding of interatomic bonding in solids is that in some instances, the type of bond allows us to explain a material's properties. For example, consider carbon, which may exist as both graphite and diamond. Whereas graphite is relatively soft and has a "greasy" feel to it, diamond is one of the hardest known materials in nature. In addition, the electrical properties of diamond and graphite are dissimilar: diamond is a poor conductor of electricity, but graphite is a reasonably good conductor. These disparities in properties are directly attributable to a type of interatomic bonding found in graphite that does not exist in diamond (see Section 12.4).

Learning Objectives

After studying this chapter, you should be able to do the following:

1. Name the two atomic models cited, and note the differences between them.
2. Describe the important quantum-mechanical principle that relates to electron energies.
3. (a) Schematically plot attractive, repulsive, and net energies versus interatomic separation for two atoms or ions.
   (b) Note on this plot the equilibrium separation and the bonding energy.
4. (a) Briefly describe ionic, covalent, metallic, hydrogen, and van der Waals bonds.
   (b) Note which materials exhibit each of these bonding types.

2.1 INTRODUCTION

Some of the important properties of solid materials depend on geometric atomic arrangements and also the interactions that exist among constituent atoms or molecules. This chapter, by way of preparation for subsequent discussions, considers several fundamental and important concepts—namely, atomic structure, electron configurations in atoms and the periodic table, and the various types of primary and secondary interatomic bonds that hold together the atoms that compose a solid. These topics are reviewed briefly, under the assumption that some of the material is familiar to the reader.

Atomic Structure

2.2 FUNDAMENTAL CONCEPTS

Each atom consists of a very small nucleus composed of protons and neutrons and is encircled by moving electrons.[1] Both electrons and protons are electrically charged, the charge magnitude being 1.602×10^{-19} C, which is negative in sign for electrons and positive for protons; neutrons are electrically neutral. Masses for these subatomic particles are extremely small; protons and neutrons have approximately the same mass, 1.67×10^{-27} kg, which is significantly larger than that of an electron, 9.11×10^{-31} kg.

atomic number (*Z*)

Each chemical element is characterized by the number of protons in the nucleus, or the **atomic number (*Z*).**[2] For an electrically neutral or complete atom, the atomic number also equals the number of electrons. This atomic number ranges in integral units from 1 for hydrogen to 92 for uranium, the highest of the naturally occurring elements.

The *atomic mass* (*A*) of a specific atom may be expressed as the sum of the masses of protons and neutrons within the nucleus. Although the number of protons is the same for

[1]Protons, neutrons, and electrons are composed of other subatomic particles such as quarks, neutrinos, and bosons. However, this discussion is concerned only with protons, neutrons, and electrons.

[2]Terms appearing in **boldface** type are defined in the Glossary, which follows Appendix E.

isotope

atomic weight

atomic mass unit (amu)

all atoms of a given element, the number of neutrons (N) may be variable. Thus atoms of some elements have two or more different atomic masses, which are called **isotopes.** The **atomic weight** of an element corresponds to the weighted average of the atomic masses of the atom's naturally occurring isotopes.[3] The **atomic mass unit (amu)** may be used to compute atomic weight. A scale has been established whereby 1 amu is defined as $\frac{1}{12}$ of the atomic mass of the most common isotope of carbon, carbon 12 (^{12}C) ($A = 12.00000$). Within this scheme, the masses of protons and neutrons are slightly greater than unity, and

$$A \cong Z + N \tag{2.1}$$

mole

The atomic weight of an element or the molecular weight of a compound may be specified on the basis of amu per atom (molecule) or mass per mole of material. In one **mole** of a substance, there are 6.022×10^{23} (Avogadro's number) atoms or molecules. These two atomic weight schemes are related through the following equation:

$$1 \text{ amu/atom (or molecule)} = 1 \text{ g/mol}$$

For example, the atomic weight of iron is 55.85 amu/atom, or 55.85 g/mol. Sometimes use of amu per atom or molecule is convenient; on other occasions, grams (or kilograms) per mole is preferred. The latter is used in this book.

EXAMPLE PROBLEM 2.1

Average Atomic Weight Computation for Cerium

Cerium has four naturally occurring isotopes: 0.185% of ^{136}Ce, with an atomic weight of 135.907 amu; 0.251% of ^{138}Ce, with an atomic weight of 137.906 amu; 88.450% of ^{140}Ce, with an atomic weight of 139.905 amu; and 11.114% of ^{142}Ce, with an atomic weight of 141.909 amu. Calculate the average atomic weight of Ce.

Solution

The average atomic weight of a hypothetical element M, $\overline{A}_M$, is computed by adding fraction-of-occurrence—atomic weight products for all its isotopes; that is,

$$\overline{A}_M = \sum_i f_{i_M} A_{i_M} \tag{2.2}$$

In this expression, f_{i_M} is the fraction-of-occurrence of isotope i for element M (i.e., the percentage-of-occurrence divided by 100), and A_{i_M} is the atomic weight of the isotope.

For cerium, Equation 2.2 takes the form

$$\overline{A}_{Ce} = f_{^{136}Ce} A_{^{136}Ce} + f_{^{138}Ce} A_{^{138}Ce} + f_{^{140}Ce} A_{^{140}Ce} + f_{^{142}Ce} A_{^{142}Ce}$$

Incorporating values provided in the problem statement for the several parameters leads to

$$\begin{aligned}\overline{A}_{Ce} &= \left(\frac{0.185\%}{100}\right)(135.907 \text{ amu}) + \left(\frac{0.251\%}{100}\right)(137.906 \text{ amu}) + \left(\frac{88.450\%}{100}\right)(139.905 \text{ amu}) \\ &\quad + \left(\frac{11.114\%}{100}\right)(141.909 \text{ amu}) \\ &= (0.00185)(135.907 \text{ amu}) + (0.00251)(137.906 \text{ amu}) + (0.8845)(139.905 \text{ amu}) \\ &\quad + (0.11114)(141.909 \text{ amu}) \\ &= 140.115 \text{ amu}\end{aligned}$$

[3]The term *atomic mass* is really more accurate than *atomic weight* inasmuch as, in this context, we are dealing with masses and not weights. However, atomic weight is, by convention, the preferred terminology and is used throughout this book. The reader should note that it is *not* necessary to divide molecular weight by the gravitational constant.

Concept Check 2.1 Why are the atomic weights of the elements generally not integers? Cite two reasons.

[*The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).*]

2.3 ELECTRONS IN ATOMS

Atomic Models

During the latter part of the nineteenth century, it was realized that many phenomena involving electrons in solids could not be explained in terms of classical mechanics. What followed was the establishment of a set of principles and laws that govern systems of atomic and subatomic entities that came to be known as **quantum mechanics.** An understanding of the behavior of electrons in atoms and crystalline solids necessarily involves the discussion of quantum-mechanical concepts. However, a detailed exploration of these principles is beyond the scope of this text, and only a very superficial and simplified treatment is given.

quantum mechanics

Bohr atomic model

One early outgrowth of quantum mechanics was the simplified **Bohr atomic model,** in which electrons are assumed to revolve around the atomic nucleus in discrete orbitals, and the position of any particular electron is more or less well defined in terms of its orbital. This model of the atom is represented in Figure 2.1.

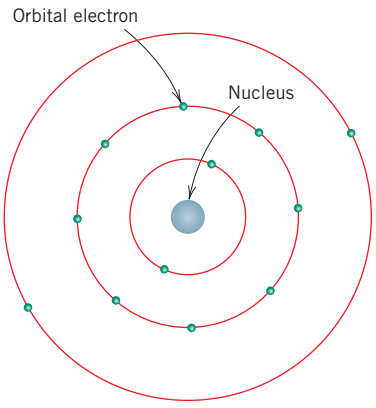
Another important quantum-mechanical principle stipulates that the energies of electrons are *quantized*—that is, electrons are permitted to have only specific values of energy. An electron may change energy, but in doing so, it must make a quantum jump either to an allowed higher energy (with absorption of energy) or to a lower energy (with emission of energy). Often, it is convenient to think of these allowed electron energies as being associated with *energy levels* or *states*. These states do not vary continuously with energy—that is, adjacent states are separated by finite energies. For example, allowed states for the Bohr hydrogen atom are represented in Figure 2.2*a*. These energies are taken to be negative, whereas the zero reference is the unbound or free electron. Of course, the single electron associated with the hydrogen atom fills only one of these states.

Thus, the Bohr model represents an early attempt to describe electrons in atoms, in terms of both position (electron orbitals) and energy (quantized energy levels).

This Bohr model was eventually found to have some significant limitations because of its inability to explain several phenomena involving electrons. A resolution was reached with a **wave-mechanical model,** in which the electron is considered to exhibit both wave-like and particle-like characteristics. With this model, an electron is no longer treated as a particle moving in a discrete orbital; rather, position is considered to be the probability of an electron being at various locations around the nucleus. In other words, position is

wave-mechanical model

Figure 2.1 Schematic representation of the Bohr atom.



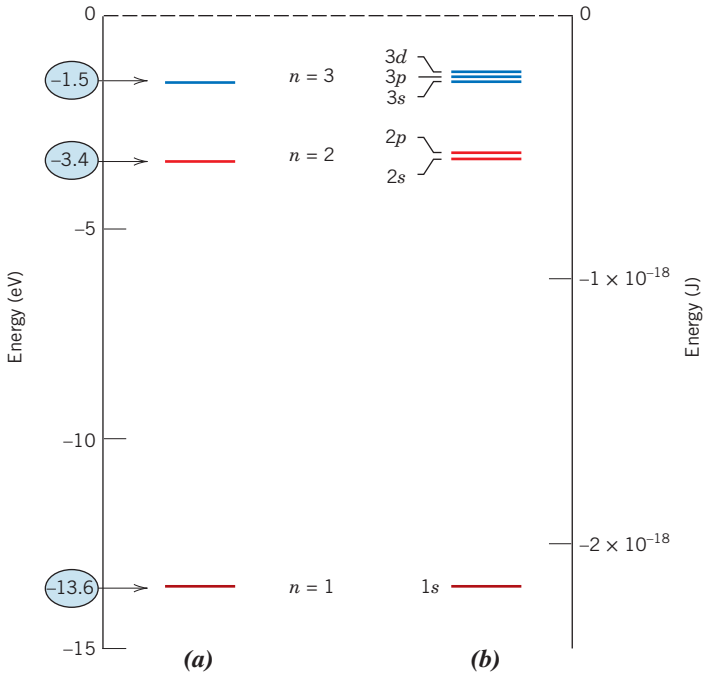


Figure 2.2 (*a*) The first three electron energy states for the Bohr hydrogen atom. (*b*) Electron energy states for the first three shells of the wave-mechanical hydrogen atom. (Adapted from W. G. Moffatt, G. W. Pearsall, and J. Wulff, *The Structure and Properties of Materials,* Vol. I, *Structure,* John Wiley & Sons, 1964. Reproduced with permission of Janet M. Moffatt.)

described by a probability distribution or electron cloud. Figure 2.3 compares Bohr and wave-mechanical models for the hydrogen atom. Both models are used throughout the course of this text; the choice depends on which model allows the simplest explanation.

Quantum Numbers

quantum number

In wave mechanics, every electron in an atom is characterized by four parameters called **quantum numbers.** The size, shape, and spatial orientation of an electron's probability density (or *orbital*) are specified by three of these quantum numbers. Furthermore, Bohr energy levels separate into electron subshells, and quantum numbers dictate the number of states within each subshell. Shells are specified by a *principal quantum number n,* which may take on integral values beginning with unity; sometimes these shells are designated by the letters *K*, *L*, *M*, *N*, *O*, and so on, which correspond, respectively,

Table 2.1 Summary of the Relationships among the Quantum Numbers *n*, *l*, m_l and Numbers of Orbitals and Electrons

Value of n	*Value of l*	*Values of* m_l	*Subshell*	*Number of Orbitals*	*Number of Electrons*
1	0	0	1*s*	1	2
2	0	0	2*s*	1	2
	1	−1, 0, +1	2*p*	3	6
3	0	0	3*s*	1	2
	1	−1, 0, +1	3*p*	3	6
	2	−2, −1, 0, +1, +2	3*d*	5	10
4	0	0	4*s*	1	2
	1	−1, 0, +1	4*p*	3	6
	2	−2, −1, 0, +1, +2	4*d*	5	10
	3	−3, −2, −1, 0, +1, +2, +3	4*f*	7	14

Source: From J. E. Brady and F. Senese, *Chemistry: Matter and Its Changes,* 4th edition, 2004. Reprinted with permission of John Wiley & Sons, Inc.

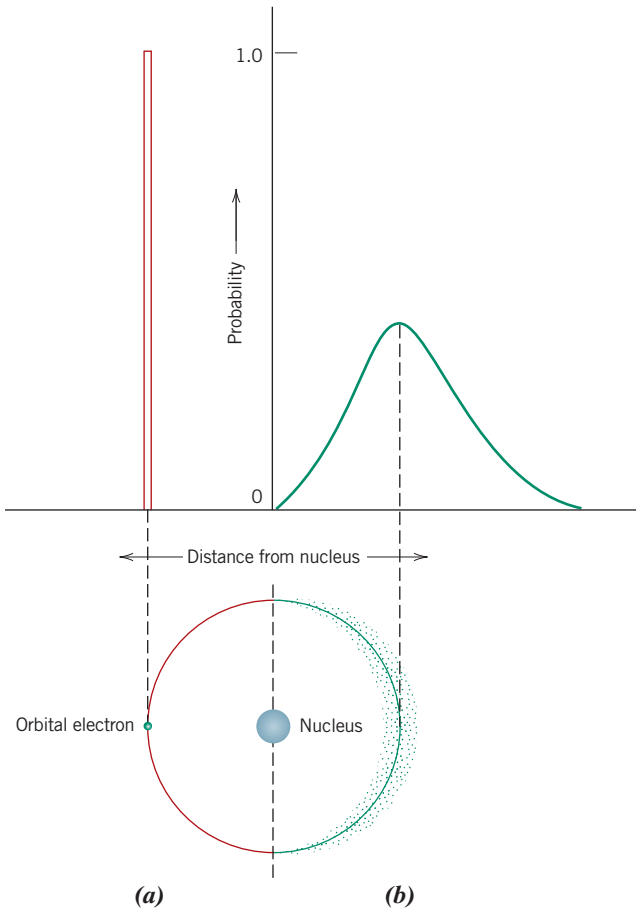


Figure 2.3 Comparison of the (*a*) Bohr and (*b*) wave-mechanical atom models in terms of electron distribution. (Adapted from Z. D. Jastrzebski, *The Nature and Properties of Engineering Materials*, 3rd edition, p. 4. Copyright © 1987 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)

to $n = 1, 2, 3, 4, 5, \ldots$, as indicated in Table 2.1. Note also that this quantum number, and it only, is also associated with the Bohr model. This quantum number is related to the size of an electron's orbital (or its average distance from the nucleus).

The second (or *azimuthal*) quantum number, l, designates the subshell. Values of l are restricted by the magnitude of n and can take on integer values that range from $l = 0$ to $l = (n - 1)$. Each subshell is denoted by a lowercase letter—an *s*, *p*, *d*, or *f*—related to l values as follows:

Value of l	*Letter Designation*
0	*s*
1	*p*
2	*d*
3	*f*

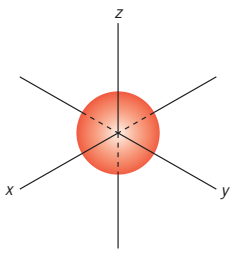


Figure 2.4 Spherical shape of an *s* electron orbital.

Furthermore, electron orbital shapes depend on l. For example *s* orbitals are spherical and centered on the nucleus (Figure 2.4). There are three orbitals for a *p* subshell (as explained next); each has a nodal surface in the shape of a dumbbell (Figure 2.5). Axes for these three orbitals are mutually perpendicular to one another like those of an *x*-*y*-*z* coordinate system; thus, it is convenient to label these orbitals p_x, p_y, and p_z (see Figure 2.5). Orbital configurations for *d* subshells are more complex and are not discussed here.

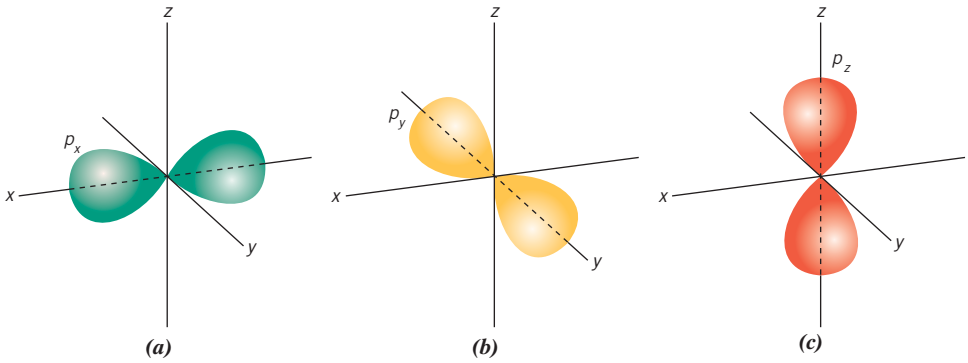


Figure 2.5 Orientations and shapes of (*a*) p_x, (*b*) p_y, and (*c*) p_z electron orbitals.

The number of electron orbitals for each subshell is determined by the third (or magnetic) quantum number, m_l; m_l can take on integer values between $-l$ and $+l$, including 0. When $l = 0$, m_l can only have a value of 0 because +0 and −0 are the same. This corresponds to an *s* subshell, which can have only one orbital. Furthermore, for $l = 1$, m_l can take on values of −1, 0, and +1, and three *p* orbitals are possible. Similarly, it can be shown that *d* subshells have five orbitals, and *f* subshells have seven. In the absence of an external magnetic field, all orbitals within each subshell are identical in energy. However, when a magnetic field is applied, these subshell states split, with each orbital assuming a slightly different energy. Table 2.1 presents a summary of the values and relationships among the n, l, and m_l quantum numbers.

Associated with each electron is a *spin moment*, which must be oriented either up or down. Related to this spin moment is the fourth quantum number, m_s, for which two values are possible: $+\frac{1}{2}$ (for spin up) and $-\frac{1}{2}$ (for spin down).

Thus, the Bohr model was further refined by wave mechanics, in which the introduction of three new quantum numbers gives rise to electron subshells within each shell. A comparison of these two models on this basis is illustrated, for the hydrogen atom, in Figures 2.2*a* and 2.2*b*.

A complete energy level diagram for the various shells and subshells using the wave-mechanical model is shown in Figure 2.6. Several features of the diagram are worth noting. First, the smaller the principal quantum number, the lower the energy level; for example, the energy of a 1*s* state is less than that of a 2*s* state, which in turn

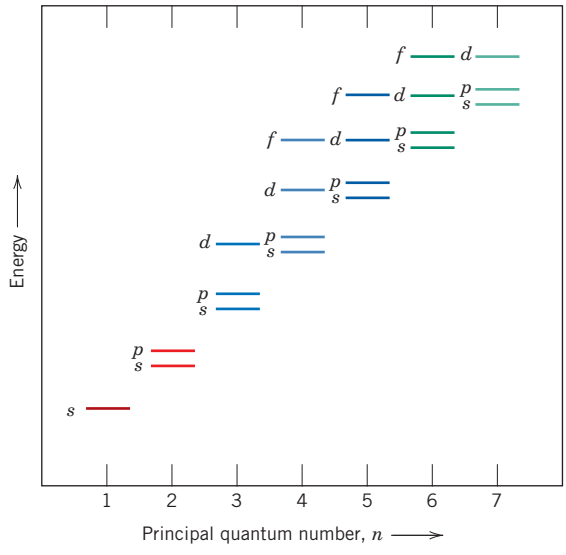


Figure 2.6 Schematic representation of the relative energies of the electrons for the various shells and subshells.
(From K. M. Ralls, T. H. Courtney, and J. Wulff, *Introduction to Materials Science and Engineering,* p. 22. Copyright © 1976 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)

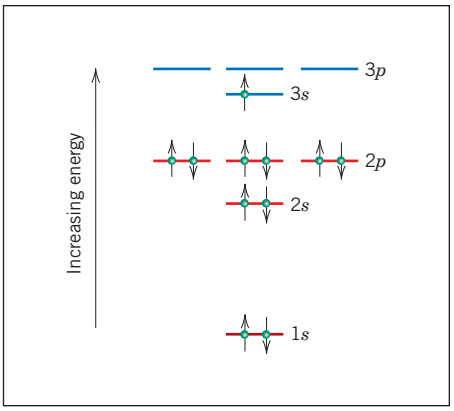


Figure 2.7 Schematic representation of the filled and lowest unfilled energy states for a sodium atom.

is lower than that of the 3*s*. Second, within each shell, the energy of a subshell level increases with the value of the *l* quantum number. For example, the energy of a 3*d* state is greater than that of a 3*p*, which is larger than 3*s*. Finally, there may be overlap in energy of a state in one shell with states in an adjacent shell, which is especially true of *d* and *f* states; for example, the energy of a 3*d* state is generally greater than that of a 4*s*.

Electron Configurations

electron state

The preceding discussion has dealt primarily with **electron states**—values of energy that are permitted for electrons. To determine the manner in which these states are filled with electrons, we use the **Pauli exclusion principle,** another quantum-mechanical concept, which stipulates that each electron state can hold no more than two electrons that must have opposite spins. Thus, *s*, *p*, *d*, and *f* subshells may each accommodate, respectively, a total of 2, 6, 10, and 14 electrons; the right column of Table 2.1 notes the maximum number of electrons that may occupy each orbital for the first four shells.

Pauli exclusion principle

Of course, not all possible states in an atom are filled with electrons. For most atoms, the electrons fill up the lowest possible energy states in the electron shells and subshells, two electrons (having opposite spins) per state. The energy structure for a sodium atom is represented schematically in Figure 2.7. When all the electrons occupy the lowest possible energies in accord with the foregoing restrictions, an atom is said to be in its **ground state.** However, electron transitions to higher energy states are possible, as discussed in Chapters 18 and 21. The **electron configuration** or structure of an atom represents the manner in which these states are occupied. In the conventional notation, the number of electrons in each subshell is indicated by a superscript after the shell–subshell designation. For example, the electron configurations for hydrogen, helium, and sodium are, respectively, $1s^1$, $1s^2$, and $1s^2 2s^2 2p^6 3s^1$. Electron configurations for some of the more common elements are listed in Table 2.2.

ground state

electron configuration

valence electron

At this point, comments regarding these electron configurations are necessary. First, the **valence electrons** are those that occupy the outermost shell. These electrons are extremely important; as will be seen, they participate in the bonding between atoms to form atomic and molecular aggregates. Furthermore, many of the physical and chemical properties of solids are based on these valence electrons.

In addition, some atoms have what are termed *stable electron configurations*—that is, the states within the outermost or valence electron shell are completely filled. Normally, this corresponds to the occupation of just the *s* and *p* states for the outermost shell by a total of eight electrons, as in neon, argon, and krypton; one exception is helium, which contains only two 1*s* electrons. These elements (Ne, Ar, Kr, and He) are the inert, or noble, gases, which are virtually unreactive chemically. Some atoms of the elements that have unfilled valence shells assume stable electron configurations by gaining or losing electrons

Table 2.2
Expected Electron Configurations for Some Common Elements[a]

Element	*Symbol*	*Atomic Number*	*Electron Configuration*
Hydrogen	H	1	$1s^1$
Helium	He	2	$1s^2$
Lithium	Li	3	$1s^2 2s^1$
Beryllium	Be	4	$1s^2 2s^2$
Boron	B	5	$1s^2 2s^2 2p^1$
Carbon	C	6	$1s^2 2s^2 2p^2$
Nitrogen	N	7	$1s^2 2s^2 2p^3$
Oxygen	O	8	$1s^2 2s^2 2p^4$
Fluorine	F	9	$1s^2 2s^2 2p^5$
Neon	Ne	10	$1s^2 2s^2 2p^6$
Sodium	Na	11	$1s^2 2s^2 2p^6 3s^1$
Magnesium	Mg	12	$1s^2 2s^2 2p^6 3s^2$
Aluminum	Al	13	$1s^2 2s^2 2p^6 3s^2 3p^1$
Silicon	Si	14	$1s^2 2s^2 2p^6 3s^2 3p^2$
Phosphorus	P	15	$1s^2 2s^2 2p^6 3s^2 3p^3$
Sulfur	S	16	$1s^2 2s^2 2p^6 3s^2 3p^4$
Chlorine	Cl	17	$1s^2 2s^2 2p^6 3s^2 3p^5$
Argon	Ar	18	$1s^2 2s^2 2p^6 3s^2 3p^6$
Potassium	K	19	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
Calcium	Ca	20	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
Scandium	Sc	21	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$
Titanium	Ti	22	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$
Vanadium	V	23	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$
Chromium	Cr	24	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
Manganese	Mn	25	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$
Iron	Fe	26	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$
Cobalt	Co	27	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$
Nickel	Ni	28	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$
Copper	Cu	29	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$
Zinc	Zn	30	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2$
Gallium	Ga	31	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^1$
Germanium	Ge	32	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^2$
Arsenic	As	33	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^3$
Selenium	Se	34	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^4$
Bromine	Br	35	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^5$
Krypton	Kr	36	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$

[a]When some elements covalently bond, they form *sp* hybrid bonds. This is especially true for C, Si, and Ge.

to form charged ions or by sharing electrons with other atoms. This is the basis for some chemical reactions and also for atomic bonding in solids, as explained in Section 2.6.

Concept Check 2.2 Give electron configurations for the Fe^{3+} and S^{2-} ions.

[*The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).*]

2.4 THE PERIODIC TABLE

periodic table

All the elements have been classified according to electron configuration in the **periodic table** (Figure 2.8). Here, the elements are situated, with increasing atomic number, in seven horizontal rows called *periods*. The arrangement is such that all elements arrayed in a given column or group have similar valence electron structures, as well as chemical and physical properties. These properties change gradually, moving horizontally across each period and vertically down each column.

The elements positioned in Group 0, the rightmost group, are the *inert gases*, which have filled electron shells and stable electron configurations.[4] Group VIIA and VIA elements are one and two electrons deficient, respectively, from having stable structures. The Group

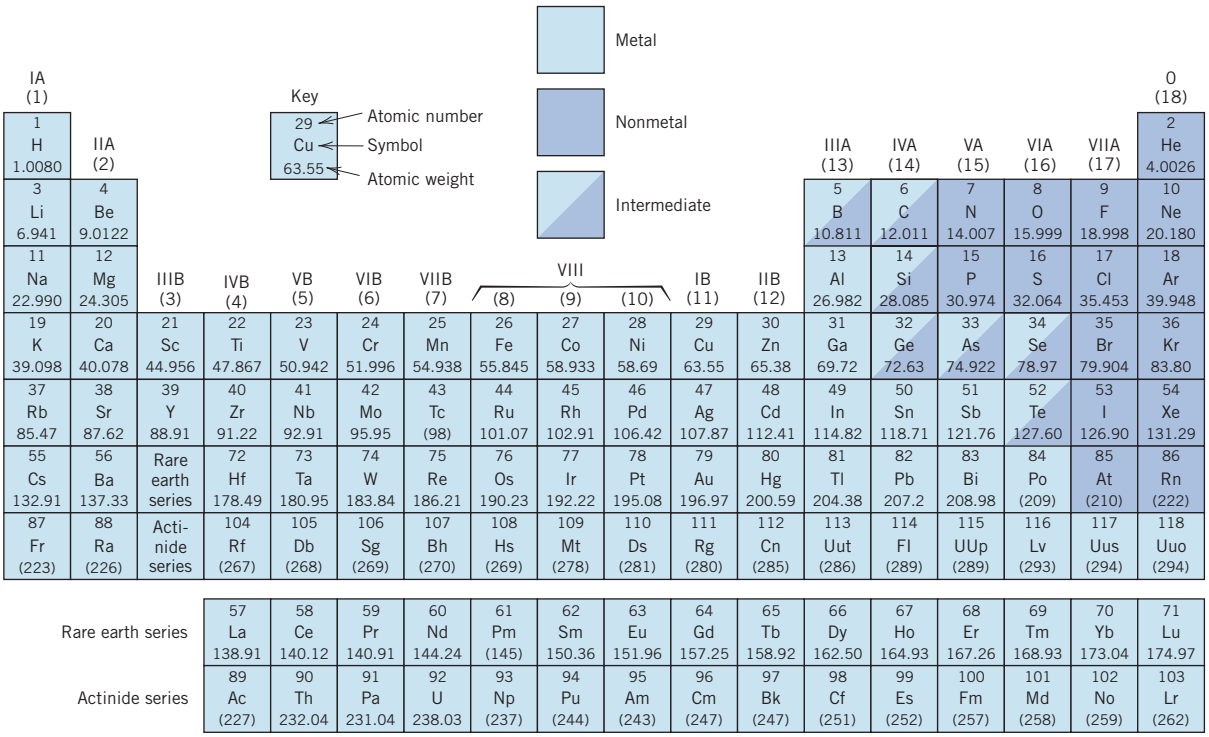


Figure 2.8 The periodic table of the elements. The numbers in parentheses are the atomic weights of the most stable or common isotopes.

[4]Two different group designation schemes are used in Figure 2.8, which appear at the top of each column. The convention used prior to 1988 was to label each group with a Roman numeral, in most instances followed by an "A" or "B". With the current system, devised by an international naming convention, the groups are numbered numerically from 1 to 18, moving from with the leftmost column (the alkali metals) to the rightmost column (the inert gases). These numbers appear in parentheses in Figure 2.8.

VIIA elements (F, Cl, Br, I, and At) are sometimes termed the *halogens*. The alkali and the alkaline earth metals (Li, Na, K, Be, Mg, Ca, etc.) are labeled as Groups IA and IIA, having, respectively, one and two electrons in excess of stable structures. The elements in the three long periods, Groups IIIB through IIB, are termed the *transition metals,* which have partially filled *d* electron states and in some cases one or two electrons in the next higher energy shell. Groups IIIA, IVA, and VA (B, Si, Ge, As, etc.) display characteristics that are intermediate between the metals and nonmetals by virtue of their valence electron structures.

electropositive

As may be noted from the periodic table, most of the elements really come under the metal classification. These are sometimes termed **electropositive** elements, indicating that they are capable of giving up their few valence electrons to become positively charged ions. Furthermore, the elements situated on the right side of the table are **electronegative**—that is, they readily accept electrons to form negatively charged ions, or sometimes they share electrons with other atoms. Figure 2.9 displays electronegativity values that have been assigned to the various elements arranged in the periodic table. As a general rule, electronegativity increases in moving from left to right and from bottom to top. Atoms are more likely to accept electrons if their outer shells are almost full and if they are less "shielded" from (i.e., closer to) the nucleus.

electronegative

In addition to chemical behavior, physical properties of the elements also tend to vary systematically with position in the periodic table. For example, most metals that reside in the center of the table (Groups IIIB through IIB) are relatively good conductors of electricity and heat; nonmetals are typically electrical and thermal insulators. Mechanically, the metallic elements exhibit varying degrees of *ductility*—the ability to be plastically deformed without fracturing (e.g., the ability to be rolled into thin sheets). Most of the nonmetals are either gases or liquids, or in the solid state are brittle in nature. Furthermore, for the Group IVA elements [C (diamond), Si, Ge, Sn, and Pb], electrical conductivity increases as we move down this column. The Group VB metals (V, Nb, and Ta) have very high melting temperatures, which increase in going down this column.

It should be noted that there is not always this consistency in property variations within the periodic table. Physical properties change in a more or less regular manner; however, there are some rather abrupt changes when one moves across a period or down a group.

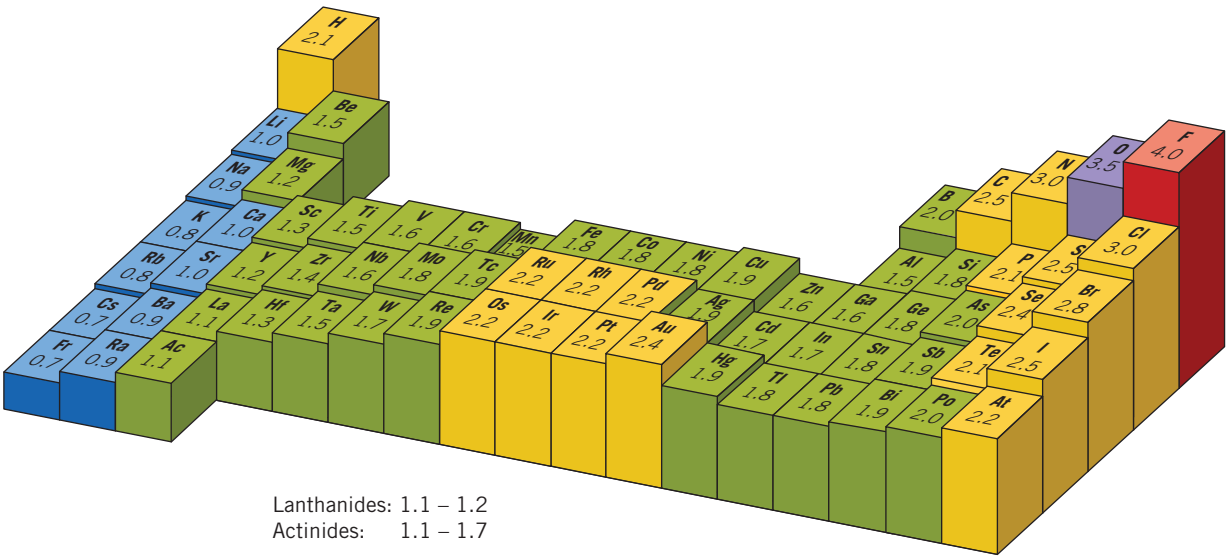


Figure 2.9 The electronegativity values for the elements.
(From J. E. Brady and F. Senese, *Chemistry: Matter and Its Changes,* 4th edition, 2004. This material is reproduced with permission of John Wiley & Sons, Inc.)

Atomic Bonding in Solids

2.5 BONDING FORCES AND ENERGIES

An understanding of many of the physical properties of materials is enhanced by a knowledge of the interatomic forces that bind the atoms together. Perhaps the principles of atomic bonding are best illustrated by considering how two isolated atoms interact as they are brought close together from an infinite separation. At large distances, interactions are negligible because the atoms are too far apart to have an influence on each other; however, at small separation distances, each atom exerts forces on the others. These forces are of two types, attractive (F_A) and repulsive (F_R), and the magnitude of each depends on the separation or interatomic distance (r); Figure 2.10*a* is a schematic plot of F_A and F_R versus r. The origin of an attractive force F_A depends on the particular type of bonding that exists between the two atoms, as discussed shortly. Repulsive forces arise from interactions between the negatively charged electron clouds for the two atoms and are important only at small values of r as the outer electron shells of the two atoms begin to overlap (Figure 2.10*a*).

The net force F_N between the two atoms is just the sum of both attractive and repulsive components; that is,

$$F_N = F_A + F_R \tag{2.3}$$

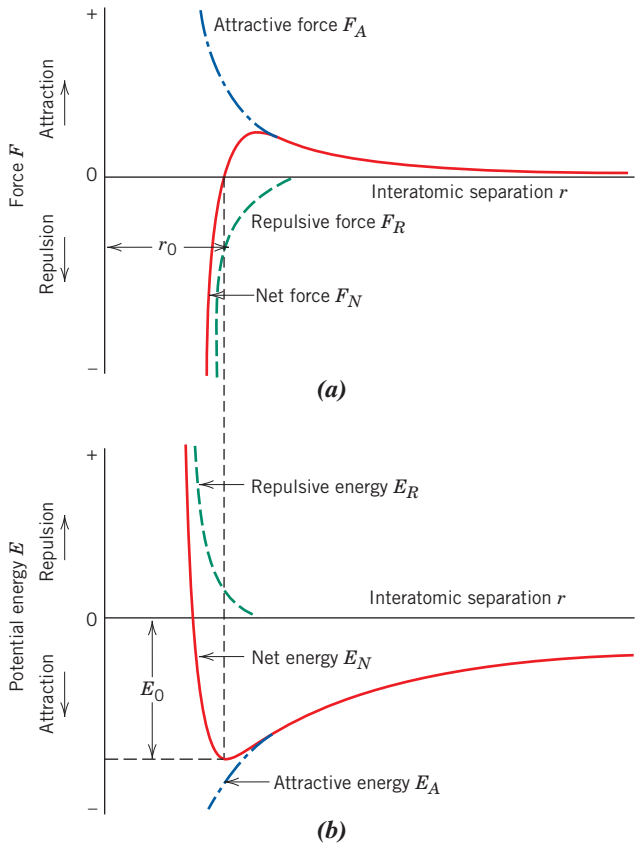


Figure 2.10 (*a*) The dependence of repulsive, attractive, and net forces on interatomic separation for two isolated atoms. (*b*) The dependence of repulsive, attractive, and net potential energies on interatomic separation for two isolated atoms.

which is also a function of the interatomic separation, as also plotted in Figure 2.10*a*. When F_A and F_R are equal in magnitude but opposite in sign, there is no net force—that is,

$$F_A + F_R = 0 \tag{2.4}$$

and a state of equilibrium exists. The centers of the two atoms remain separated by the equilibrium spacing r_0, as indicated in Figure 2.10*a*. For many atoms, r_0 is approximately 0.3 nm. Once in this position, any attempt to move the two atoms farther apart is counteracted by the attractive force, while pushing them closer together is resisted by the increasing repulsive force.

Sometimes it is more convenient to work with the potential energies between two atoms instead of forces. Mathematically, energy (E) and force (F) are related as

Force–potential energy relationship for two atoms

$$E = \int F\,dr \tag{2.5a}$$

And, for atomic systems,

$$E_N = \int_r^\infty F_N\,dr \tag{2.6}$$

$$= \int_r^\infty F_A\,dr + \int_r^\infty F_R\,dr \tag{2.7}$$

$$= E_A + E_R \tag{2.8a}$$

in which E_N, E_A, and E_R are, respectively, the net, attractive, and repulsive energies for two isolated and adjacent atoms.[5]

Figure 2.10*b* plots attractive, repulsive, and net potential energies as a function of interatomic separation for two atoms. From Equation 2.8a, the net curve is the sum of the attractive and repulsive curves. The minimum in the net energy curve corresponds to the equilibrium spacing, r_0. Furthermore, the **bonding energy** for these two atoms, E_0, corresponds to the energy at this minimum point (also shown in Figure 2.10*b*); it represents the energy required to separate these two atoms to an infinite separation.

bonding energy

Although the preceding treatment deals with an ideal situation involving only two atoms, a similar yet more complex condition exists for solid materials because force and energy interactions among atoms must be considered. Nevertheless, a bonding energy, analogous to E_0 above, may be associated with each atom. The magnitude of this bonding energy and the shape of the energy–versus–interatomic separation curve vary from material to material, and they both depend on the type of atomic bonding. Furthermore, a number of material properties depend on E_0, the curve shape, and bonding type.

[5]Force in Equation 2.5a may also be expressed as

$$F = \frac{dE}{dr} \tag{2.5b}$$

Likewise, the force equivalent of Equation 2.8a is as follows:

$$F_N = F_A + F_R \tag{2.3}$$

$$= \frac{dE_A}{dr} + \frac{dE_R}{dr} \tag{2.8b}$$

For example, materials having large bonding energies typically also have high melting temperatures; at room temperature, solid substances are formed for large bonding energies, whereas for small energies, the gaseous state is favored; liquids prevail when the energies are of intermediate magnitude. In addition, as discussed in Section 6.3, the mechanical stiffness (or modulus of elasticity) of a material is dependent on the shape of its force–versus–interatomic separation curve (Figure 6.7). The slope for a relatively stiff material at the $r = r_0$ position on the curve will be quite steep; slopes are shallower for more flexible materials. Furthermore, how much a material expands upon heating or contracts upon cooling (i.e., its linear coefficient of thermal expansion) is related to the shape of its E–versus–r curve (see Section 19.3). A deep and narrow "trough," which typically occurs for materials having large bonding energies, normally correlates with a low coefficient of thermal expansion and relatively small dimensional alterations for changes in temperature.

primary bond

Three different types of **primary** or chemical **bond** are found in solids—ionic, covalent, and metallic. For each type, the bonding necessarily involves the valence electrons; furthermore, the nature of the bond depends on the electron structures of the constituent atoms. In general, each of these three types of bonding arises from the tendency of the atoms to assume stable electron structures, like those of the inert gases, by completely filling the outermost electron shell.

Secondary or physical forces and energies are also found in many solid materials; they are weaker than the primary ones but nonetheless influence the physical properties of some materials. The sections that follow explain the several kinds of primary and secondary interatomic bonds.

2.6 PRIMARY INTERATOMIC BONDS

Ionic Bonding

ionic bonding

Ionic bonding is perhaps the easiest to describe and visualize. It is always found in compounds composed of both metallic and nonmetallic elements, elements situated at the horizontal extremities of the periodic table. Atoms of a metallic element easily give up their valence electrons to the nonmetallic atoms. In the process, all the atoms acquire stable or inert gas configurations (i.e., completely filled orbital shells) and, in addition, an electrical charge—that is, they become ions. Sodium chloride (NaCl) is the classic ionic material. A sodium atom can assume the electron structure of neon (and a net single positive charge with a reduction in size) by a transfer of its one valence 3*s* electron to a chlorine atom (Figure 2.11*a*). After such a transfer, the chlorine ion acquires a net negative charge, an electron configuration identical to that of argon; it is also larger than the chlorine atom. Ionic bonding is illustrated schematically in Figure 2.11*b*.

coulombic force

The attractive bonding forces are **coulombic**—that is, positive and negative ions, by virtue of their net electrical charge, attract one another. For two isolated ions, the attractive energy E_A is a function of the interatomic distance according to

Attractive energy—interatomic separation relationship

$$E_A = -\frac{A}{r} \tag{2.9}$$

Theoretically, the constant A is equal to

$$A = \frac{1}{4\pi\varepsilon_0}(|Z_1|e)(|Z_2|e) \tag{2.10}$$

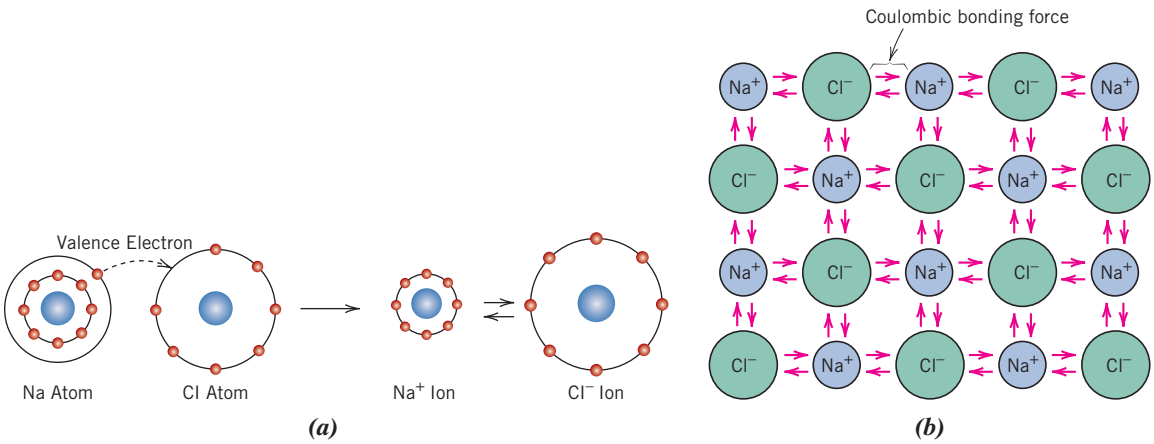


Figure 2.11 Schematic representations of (*a*) the formation of Na^+ and Cl^- ions and (*b*) ionic bonding in sodium chloride (NaCl).

Here ε_0 is the permittivity of a vacuum (8.85×10^{-12} F/m), $|Z_1|$ and $|Z_2|$ are absolute values of the valences for the two ion types, and e is the electronic charge (1.602×10^{-19} C). The value of A in Equation 2.9 assumes the bond between ions 1 and 2 is totally ionic (see Equation 2.16). Inasmuch as bonds in most of these materials are not 100% ionic, the value of A is normally determined from experimental data rather than computed using Equation 2.10.

An analogous equation for the repulsive energy is[6]

Repulsive energy—interatomic separation relationship

$$E_R = \frac{B}{r^n} \tag{2.11}$$

In this expression, B and n are constants whose values depend on the particular ionic system. The value of n is approximately 8.

Ionic bonding is termed *nondirectional*—that is, the magnitude of the bond is equal in all directions around an ion. It follows that for ionic materials to be stable, all positive ions must have as nearest neighbors negatively charged ions in a three-dimensional scheme, and vice versa. Some of the ion arrangements for these materials are discussed in Chapter 12.

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Tutorial Video: Bonding

What Is Ionic Bonding?

Bonding energies, which generally range between 600 and 1500 kJ/mol, are relatively large, as reflected in high melting temperatures.[7] Table 2.3 contains bonding energies and melting temperatures for several ionic materials. Interatomic bonding is typified by ceramic materials, which are characteristically hard and brittle and, furthermore, electrically and thermally insulative. As discussed in subsequent chapters, these properties are a direct consequence of electron configurations and/or the nature of the ionic bond.

[6]In Equation 2.11, the value of the constant B is also fit using experimental data.

[7]Sometimes bonding energies are expressed per atom or per ion. Under these circumstances, the electron volt (eV) is a conveniently small unit of energy. It is, by definition, the energy imparted to an electron as it falls through an electric potential of one volt. The joule equivalent of the electron volt is as follows: 1.602×10^{-19} J = 1 eV.

Table 2.3
Bonding Energies and Melting Temperatures for Various Substances

Substance	*Bonding Energy (kJ/mol)*	*Melting Temperature (°C)*
	Ionic	
NaCl	640	801
LiF	850	848
MgO	1000	2800
CaF_2	1548	1418
	Covalent	
Cl_2	121	−102
Si	450	1410
InSb	523	942
C (diamond)	713	>3550
SiC	1230	2830
	Metallic	
Hg	62	−39
Al	330	660
Ag	285	962
W	850	3414
	van der Waals[a]	
Ar	7.7	−189 (@ 69 kPa)
Kr	11.7	−158 (@ 73.2 kPa)
CH_4	18	−182
Cl_2	31	−101
	Hydrogen[a]	
HF	29	−83
NH_3	35	−78
H_2O	51	0

[a]Values for van der Waals and hydrogen bonds are energies *between* molecules or atoms (*inter*molecular), not between atoms within a molecule (*intra*molecular).

EXAMPLE PROBLEM 2.2

Computation of Attractive and Repulsive Forces between Two Ions

The atomic radii of K^+ and Br^- ions are 0.138 and 0.196 nm, respectively.

(a) Using Equations 2.9 and 2.10, calculate the force of attraction between these two ions at their equilibrium interionic separation (i.e., when the ions just touch one another).
(b) What is the force of repulsion at this same separation distance?

Solution

(a) From Equation 2.5b, the force of attraction between two ions is

$$F_A = \frac{dE_A}{dr}$$

Whereas, according to Equation 2.9,

$$E_A = -\frac{A}{r}$$

Now, taking the derivation of E_A with respect to r yields the following expression for the force of attraction F_A:

$$F_A = \frac{dE_A}{dr} = \frac{d\left(-\frac{A}{r}\right)}{dr} = -\left(\frac{-A}{r^2}\right) = \frac{A}{r^2} \tag{2.12}$$

Now substitution into this equation the expression for A (Equation 2.10) gives

$$F_A = \frac{1}{4\pi\varepsilon_0 r^2}(|Z_1|e)(|Z_2|e) \tag{2.13}$$

Incorporation into this equation values for e and ε_0 leads to

$$\begin{aligned} F_A &= \frac{1}{4\pi(8.85 \times 10^{-12}\ \text{F/m})(r^2)}[|Z_1|(1.602 \times 10^{-19}\ \text{C})][|Z_2|(1.602 \times 10^{-19}\ \text{C})] \\ &= \frac{(2.31 \times 10^{-28}\ \text{N}\cdot\text{m}^2)(|Z_1|)(|Z_2|)}{r^2} \end{aligned} \tag{2.14}$$

For this problem, r is taken as the interionic separation r_0 for KBr, which is equal to the sum of the K^+ and Br^- ionic radii inasmuch as the ions touch one another—that is,

$$\begin{aligned} r_0 &= r_{K^+} + r_{Br^-} \\ &= 0.138\ \text{nm} + 0.196\ \text{nm} \\ &= 0.334\ \text{nm} \\ &= 0.334 \times 10^{-9}\ \text{m} \end{aligned} \tag{2.15}$$

When we substitute this value for r into Equation 2.14, and taking ion 1 to be K^+ and ion 2 as Br^- (i.e., $Z_1 = +1$ and $Z_2 = -1$), then the force of attraction is equal to

$$F_A = \frac{(2.31 \times 10^{-28}\ \text{N}\cdot\text{m}^2)(|+1|)(|-1|)}{(0.334 \times 10^{-9}\ \text{m})^2} = 2.07 \times 10^{-9}\ \text{N}$$

(b) At the equilibrium separation distance the sum of attractive and repulsive forces is zero according to Equation 2.4. This means that

$$F_R = -F_A = -(2.07 \times 10^{-9}\ \text{N}) = -2.07 \times 10^{-9}\ \text{N}$$

Covalent Bonding

covalent bonding

A second bonding type, **covalent bonding,** is found in materials whose atoms have small differences in electronegativity—that is, that lie near one another in the periodic table. For these materials, stable electron configurations are assumed by the sharing of electrons between adjacent atoms. Two covalently bonded atoms will each contribute at least one electron to the bond, and the shared electrons may be considered to belong to both atoms. Covalent bonding is schematically illustrated in Figure 2.12 for a molecule of hydrogen (H_2). The hydrogen atom has a single 1*s* electron. Each of the atoms can acquire a helium electron configuration (two 1*s* valence electrons) when they share their single electron (right side of Figure 2.12). Furthermore, there is an overlapping of electron orbitals in the region between the two bonding atoms. In addition, the covalent

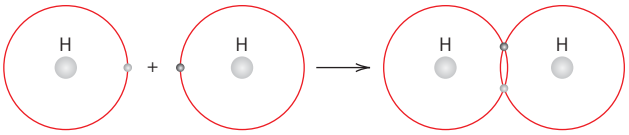


Figure 2.12 Schematic representation of covalent bonding in a molecule of hydrogen (H_2).

bond is *directional*—that is, it is between specific atoms and may exist only in the direction between one atom and another that participates in the electron sharing.

Many nonmetallic elemental molecules (e.g., Cl_2, F_2), as well as molecules containing dissimilar atoms, such as CH_4, H_2O, HNO_3, and HF, are covalently bonded.[8] Furthermore, this type of bonding is found in elemental solids such as diamond (carbon), silicon, and germanium and other solid compounds composed of elements that are located on the right side of the periodic table, such as gallium arsenide (GaAs), indium antimonide (InSb), and silicon carbide (SiC).

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Tutorial Video: Bonding
What Is Covalent Bonding?

Covalent bonds may be very strong, as in diamond, which is very hard and has a very high melting temperature, >3550°C (6400°F), or they may be very weak, as with bismuth, which melts at about 270°C (518°F). Bonding energies and melting temperatures for a few covalently bonded materials are presented in Table 2.3. Inasmuch as electrons participating in covalent bonds are tightly bound to the bonding atoms, most covalently bonded materials are electrical insulators, or, in some cases, semiconductors. Mechanical behaviors of these materials vary widely: some are relatively strong, others are weak; some fail in a brittle manner, whereas others experience significant amounts of deformation before failure. It is difficult to predict the mechanical properties of covalently bonded materials on the basis of their bonding characteristics.

Bond Hybridization in Carbon

Often associated with the covalent bonding of carbon (as well other nonmetallic substances) is the phenomenon of *hybridization*—the mixing (or combining) of two or more atomic orbitals with the result that more orbital overlap during bonding results. For example, consider the electron configuration of carbon: $1s^2 2s^2 2p^2$. Under some circumstances, one of the 2*s* orbitals is promoted to the empty 2*p* orbital (Figure 2.13*a*), which gives rise to a $1s^2 2s^1 2p^3$ configuration (Figure 2.13*b*). Furthermore, the 2*s* and 2*p* orbitals can mix to produce four sp^3 orbitals that are equivalent to one another, have parallel spins, and are capable of covalently bonding with other atoms. This orbital mixing is termed *hybridization*, which leads to the electron configuration shown in Figure 2.13*c*; here, each sp^3 orbital contains one electron, and, therefore, is half-filled.

Bonding hybrid orbitals are directional in nature—that is, each extends to and overlaps the orbital of an adjacent bonding atom. Furthermore, for carbon, each of its four sp^3 hybrid orbitals is directed symmetrically from a carbon atom to the vertex of a tetrahedron—a configuration represented schematically in Figure 2.14; the angle between each set of adjacent bonds is 109.5°.[9] The bonding of sp^3 hybrid orbitals to the 1*s* orbitals of four hydrogen atoms, as in a molecule of methane (CH_4), is presented in Figure 2.15.

For diamond, its carbon atoms are bonded to one another with sp^3 covalent hybrids—each atom is bonded to four other carbon atoms. The crystal structure for diamond is shown in Figure 12.16. Diamond's carbon–carbon bonds are extremely strong, which accounts for its high melting temperature and ultrahigh hardness (it is the hardest of all materials). Many polymeric materials are composed of long chains of carbon atoms that are also bonded together using sp^3 tetrahedral bonds; these chains form a zigzag structure (Figure 14.1*b*) because of this 109.5° interbonding angle.

[8]For these substances, the *intra*molecular bonds (bonds between atoms in a molecule) are covalent. As noted in the next section, other types of bonds can operate between molecules, which are termed *inter*molecular.

[9]Bonding of this type (to four other atoms) is sometimes termed *tetrahedral bonding*.

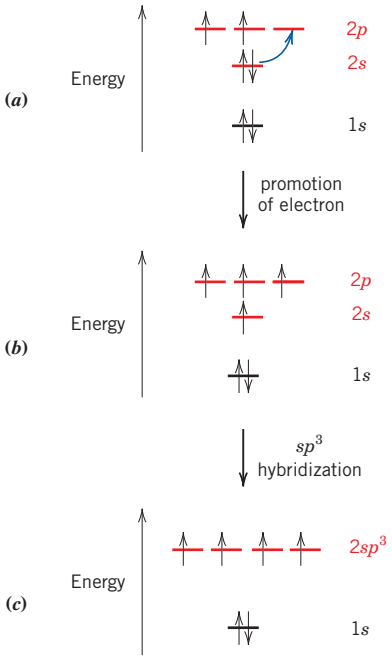


Figure 2.13 Schematic diagram that shows the formation of sp^3 hybrid orbitals in carbon. (*a*) Promotion of a 2*s* electron to a 2*p* state; (*b*) this promoted electron in a 2*p* state; (*c*) four $2sp^3$ orbitals that form by mixing the single 2*s* orbital with the three 2*p* orbitals.

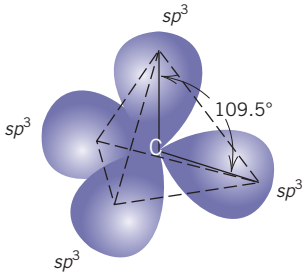


Figure 2.14 Schematic diagram showing four sp^3 hybrid orbitals that point to the corners of a tetrahedron; the angle between orbitals is 109.5°.
(From J. E. Brady and F. Senese, *Chemistry: Matter and Its Changes,* 4th edition, 2004. Reprinted with permission of John Wiley & Sons, Inc.)

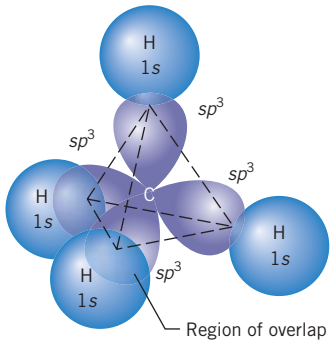


Figure 2.15 Schematic diagram that shows bonding of carbon sp^3 hybrid orbitals to the 1*s* orbitals of four hydrogen atoms in a molecule of methane (CH_4).
(From J. E. Brady and F. Senese, *Chemistry: Matter and Its Changes,* 4th edition, 2004. Reprinted with permission of John Wiley & Sons, Inc.)

Other types of hybrid bonds are possible for carbon, as well as other substances. One of these is sp^2, in which an *s* orbital and two *p* orbitals are hybridized. To achieve this configuration, one 2*s* orbital mixes with two of the three 2*p* orbitals—the third *p* orbital remains unhybridized; this is shown in Figure 2.16. Here, $2p_z$ denotes the unhybridized *p* orbital.[10] Three sp^2 hybrids belong to each carbon atom, which lie in the same plane such that the angle between adjacent orbitals is 120° (Figure 2.17); lines drawn from one orbital to another form a triangle. Furthermore, the unhybridized $2p_z$ orbital is oriented perpendicular to the plane containing the sp^2 hybrids.

These sp^2 bonds are found in graphite, another form of carbon, which has a structure and properties distinctly different from those of diamond (as discussed in Section 12.4). Graphite is composed of parallel layers of interconnecting hexagons. Hexagons form from planar sp^2 triangles that bond to one another in the manner presented in Figure 2.18—a carbon atom is located at each vertex. In-plane sp^2 bonds are strong; by way of contrast, weak interplanar bonding results from van der Waals forces that involve electrons originating from the unhybridized $2p_z$ orbitals. The structure of graphite is shown in Figure 12.17.

[10]This $2p_z$ orbital has the shape and orientation of the p_z shown in Figure 2.5*c*. In addition, the two *p* orbitals found in the sp^2 hybrid correspond to the p_x and p_y orbitals of this same figure. Furthermore, p_x, p_y, and p_z are the three orbitals of the sp^3 hybrid.

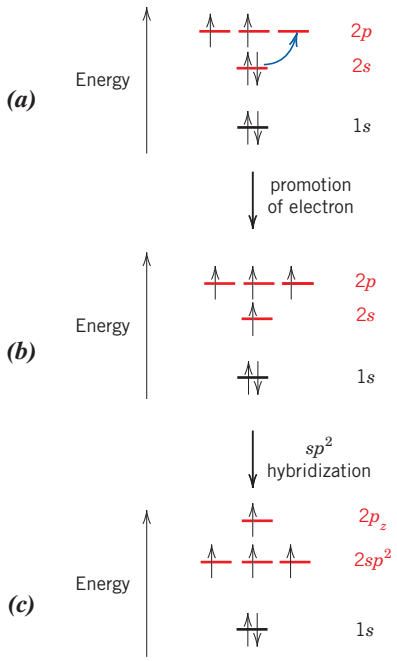


Figure 2.16 Schematic diagram that shows the formation of sp^2 hybrid orbitals in carbon. (*a*) Promotion of a 2*s* electron to a 2*p* state; (*b*) this promoted electron in a 2*p* state; (*c*) three $2sp^2$ orbitals that form by mixing the single 2*s* orbital with two 2*p* orbitals—the $2p_z$ orbital remains unhybridized.

Metallic Bonding

metallic bonding

Metallic bonding, the final primary bonding type, is found in metals and their alloys. A relatively simple model has been proposed that very nearly approximates the bonding scheme. With this model, these valence electrons are not bound to any particular atom in the solid and are more or less free to drift throughout the entire metal. They may be thought of as belonging to the metal as a whole, or forming a "sea of electrons" or an "electron cloud." The remaining nonvalence electrons and atomic nuclei form what are called *ion cores*, which possess a net positive charge equal in magnitude to the total valence electron charge

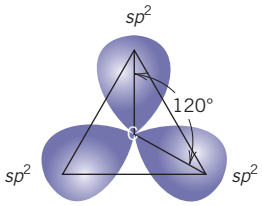


Figure 2.17 Schematic diagram showing three sp^2 orbitals that are coplanar and point to the corners of a triangle; the angle between adjacent orbitals is 120°.
(From J. E. Brady and F. Senese, *Chemistry: Matter and Its Changes,* 4th edition, 2004. Reprinted with permission of John Wiley & Sons, Inc.)

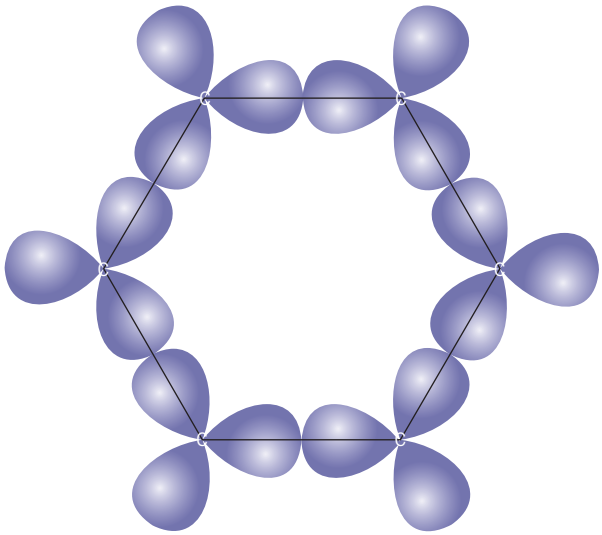


Figure 2.18 The formation of a hexagon by the bonding of six sp^2 triangles to one another.

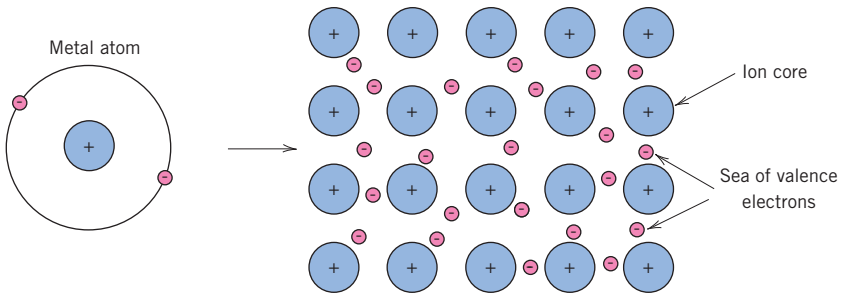


Figure 2.19 Schematic illustration of metallic bonding.

per atom. Figure 2.19 illustrates metallic bonding. The free electrons shield the positively charged ion cores from the mutually repulsive electrostatic forces that they would otherwise exert upon one another; consequently, the metallic bond is nondirectional in character. In addition, these free electrons act as a "glue" to hold the ion cores together. Bonding energies and melting temperatures for several metals are listed in Table 2.3. Bonding may be weak or strong; energies range from 62 kJ/mol for mercury to 850 kJ/mol for tungsten. Their respective melting temperatures are −39°C and 3414°C (−39°F and 6177°F).

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Tutorial Video: Bonding
What Is Metallic Bonding?

Metallic bonding is found in the periodic table for Group IA and IIA elements and, in fact, for all elemental metals.

Metals are good conductors of both electricity and heat as a consequence of their free electrons (see Sections 18.5, 18.6, and 19.4). Furthermore, in Section 7.4, we note that at room temperature, most metals and their alloys fail in a ductile manner—that is, fracture occurs after the materials have experienced significant degrees of permanent deformation. This behavior is explained in terms of a deformation mechanism (Section 7.2), which is implicitly related to the characteristics of the metallic bond.

Concept Check 2.3 Explain why covalently bonded materials are generally less dense than ionically or metallically bonded ones.

[*The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).*]

2.7 SECONDARY BONDING OR VAN DER WAALS BONDING

secondary bond
van der Waals bond

Secondary bonds, or **van der Waals** (physical) **bonds,** are weak in comparison to the primary or chemical bonds; bonding energies range between about 4 and 30 kJ/mol. Secondary bonding exists between virtually all atoms or molecules, but its presence may be obscured if any of the three primary bonding types is present. Secondary bonding is evidenced for the inert gases, which have stable electron structures. In addition, secondary (or *inter*molecular) bonds are possible between atoms or groups of atoms, which themselves are joined together by primary (or *intra*molecular) ionic or covalent bonds.

dipole

Secondary bonding forces arise from atomic or molecular **dipoles.** In essence, an electric dipole exists whenever there is some separation of positive and negative portions of an atom or molecule. The bonding results from the coulombic attraction between the positive end of one dipole and the negative region of an adjacent one, as indicated in Figure 2.20. Dipole interactions occur between induced dipoles, between induced dipoles and polar molecules (which have permanent dipoles), and between polar molecules. **Hydrogen bonding,** a special type of secondary bonding, is found to exist

hydrogen bonding

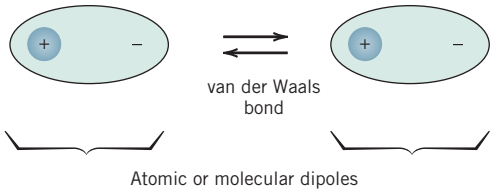


Figure 2.20 Schematic illustration of van der Waals bonding between two dipoles.

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What Is a Dipole?

between some molecules that have hydrogen as one of the constituents. These bonding mechanisms are discussed briefly next.

Fluctuating Induced Dipole Bonds

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What Is van der Waals Bonding?

A dipole may be created or induced in an atom or molecule that is normally electrically symmetric—that is, the overall spatial distribution of the electrons is symmetric with respect to the positively charged nucleus, as shown in Figure 2.21*a*. All atoms experience constant vibrational motion that can cause instantaneous and short-lived distortions of this electrical symmetry for some of the atoms or molecules and the creation of small electric dipoles. One of these dipoles can in turn produce a displacement of the electron distribution of an adjacent molecule or atom, which induces the second one also to become a dipole that is then weakly attracted or bonded to the first (Figure 2.21*b*); this is one type of van der Waals bonding. These attractive forces, which are temporary and fluctuate with time, may exist between large numbers of atoms or molecules.

The liquefaction and, in some cases, the solidification of the inert gases and other electrically neutral and symmetric molecules such as H_2 and Cl_2 are realized because of this type of bonding. Melting and boiling temperatures are extremely low in materials for which induced dipole bonding predominates; of all possible intermolecular bonds, these are the weakest. Bonding energies and melting temperatures for argon, krypton, methane, and chlorine are also tabulated in Table 2.3.

Polar Molecule–Induced Dipole Bonds

polar molecule

Permanent dipole moments exist in some molecules by virtue of an asymmetrical arrangement of positively and negatively charged regions; such molecules are termed **polar molecules.** Figure 2.22*a* shows a schematic representation of a hydrogen chloride molecule; a permanent dipole moment arises from net positive and negative charges that are respectively associated with the hydrogen and chlorine ends of the HCl molecule.

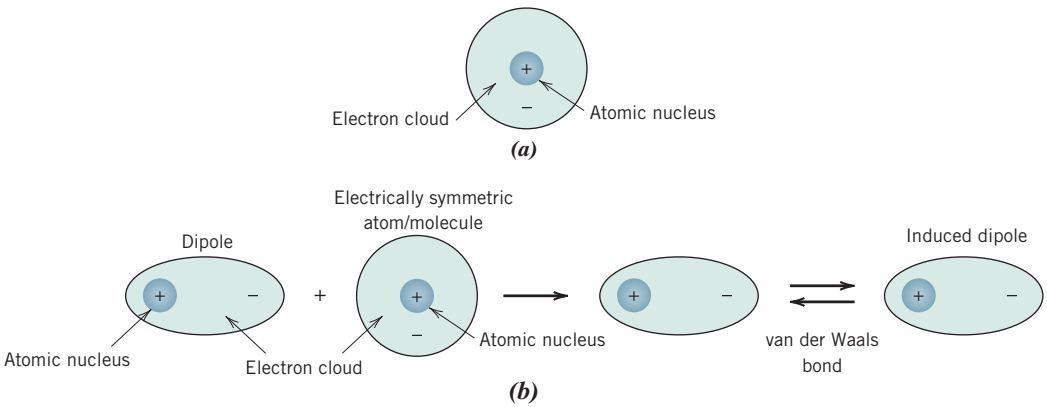


Figure 2.21 Schematic representations of (*a*) an electrically symmetric atom and (*b*) how an electric dipole induces an electrically symmetric atom/molecule to become a dipole—also the van der Waals bond between the dipoles.

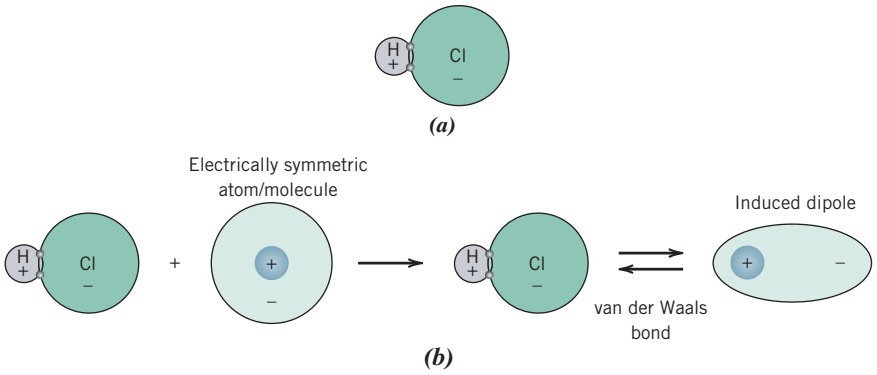


Figure 2.22 Schematic representations of (*a*) a hydrogen chloride molecule (dipole) and (*b*) how an HCl molecule induces an electrically symmetric atom/molecule to become a dipole—also the van der Waals bond between these dipoles.

Polar molecules can also induce dipoles in adjacent nonpolar molecules, and a bond forms as a result of attractive forces between the two molecules; this bonding scheme is represented schematically in Figure 2.22*b*. Furthermore, the magnitude of this bond is greater than for fluctuating induced dipoles.

Permanent Dipole Bonds

Coulombic forces also exist between adjacent polar molecules as in Figure 2.20. The associated bonding energies are significantly greater than for bonds involving induced dipoles.

The strongest secondary bonding type, the hydrogen bond, is a special case of polar molecule bonding. It occurs between molecules in which hydrogen is covalently bonded to fluorine (as in HF), oxygen (as in H_2O), or nitrogen (as in NH_3). For each H—F, H—O, or H—N bond, the single hydrogen electron is shared with the other atom. Thus, the hydrogen end of the bond is essentially a positively charged bare proton unscreened by any electrons. This highly positively charged end of the molecule is capable of a strong attractive force with the negative end of an adjacent molecule, as demonstrated in Figure 2.23 for HF. In essence, this single proton forms a bridge between two negatively charged atoms. The magnitude of the hydrogen bond is generally greater than that of the other types of secondary bonds and may be as high as 51 kJ/mol, as shown in Table 2.3. Melting and boiling temperatures for hydrogen fluoride, ammonia, and water are abnormally high in light of their low molecular weights, as a consequence of hydrogen bonding.

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Tutorial Video: Bonding
What Are the Differences between Ionic, Covalent, Metallic, and van der Waals Types of Bonding?

In spite of the small energies associated with secondary bonds, they nevertheless are involved in a number of natural phenomena and many products that we use on a daily basis. Examples of physical phenomena include the solubility of one substance in another, surface tension and capillary action, vapor pressure, volatility, and viscosity. Common applications that make use of these phenomena include *adhesives*—van der Waals bonds form between two surfaces so that they adhere to one another (as discussed in the chapter opener for this chapter); *surfactants*—compounds that lower the surface tension of a liquid and are found in soaps, detergents, and foaming agents; *emulsifiers*—substances that, when added to two immiscible materials (usually liquids), allow particles of one material to be suspended in another (common emulsions include sunscreens, salad dressings, milk, and mayonnaise); and *desiccants*—materials that form hydrogen bonds with water molecules (and remove moisture from closed containers—e.g., small packets that are often found in cartons of packaged goods); and finally, the strengths, stiffnesses, and softening temperatures of polymers, to some degree, depend on secondary bonds that form between chain molecules.

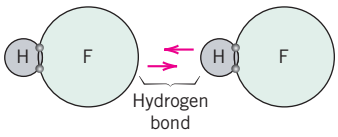


Figure 2.23 Schematic representation of hydrogen bonding in hydrogen fluoride (HF).

MATERIAL OF IMPORTANCE 2.1

Water (Its Volume Expansion Upon Freezing)

Upon freezing (i.e., transforming from a liquid to a solid upon cooling), most substances experience an increase in density (or, correspondingly, a decrease in volume). One exception is water, which exhibits the anomalous and familiar expansion upon freezing—approximately 9 volume percent expansion. This behavior may be explained on the basis of hydrogen bonding. Each H_2O molecule has two hydrogen atoms that can bond to oxygen atoms; in addition, its single O atom can bond to two hydrogen atoms of other H_2O molecules. Thus, for solid ice, each water molecule participates in four hydrogen bonds, as shown in the three-dimensional schematic of Figure 2.24*a;* here, hydrogen bonds are denoted by dashed lines, and each water molecule has 4 nearest-neighbor molecules. This is a relatively open structure—that is, the molecules are not closely packed together—and as a result, the density is comparatively low. Upon melting, this structure is partially destroyed, such that the water molecules become more closely packed together (Figure 2.24*b*)—at room temperature, the average number of nearest-neighbor water molecules has increased to approximately 4.5; this leads to an increase in density.

Consequences of this anomalous freezing phenomenon are familiar; it explains why icebergs float; why, in cold climates, it is necessary to add antifreeze to an automobile's cooling system (to keep the engine block from cracking); and why freeze–thaw cycles break up the pavement in streets and cause potholes to form.



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A watering can that ruptured along a side panel—bottom panel seam. Water that was left in the can during a cold late-autumn night expanded as it froze and caused the rupture.

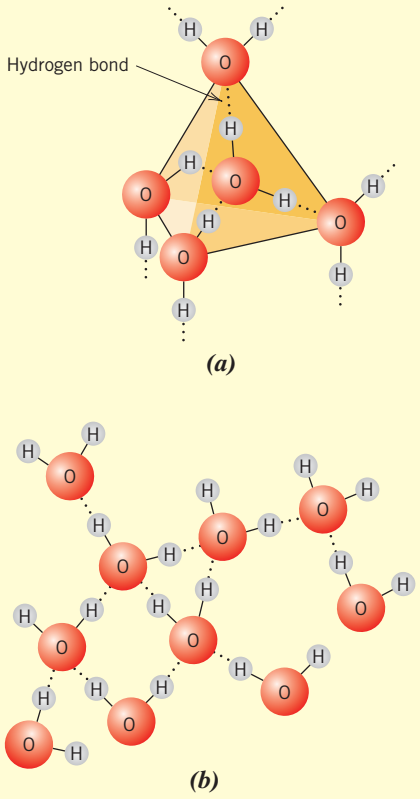


Figure 2.24 The arrangement of water (H_2O) molecules in (*a*) solid ice and (*b*) liquid water.

2.8 MIXED BONDING

Sometimes it is illustrative to represent the four bonding types—ionic, covalent, metallic, and van der Waals—on what is called a *bonding tetrahedron*—a three-dimensional tetrahedron with one of these "extreme" types located at each vertex, as shown in Figure 2.25*a*. Furthermore, we should point out that for many real materials, the atomic bonds are mixtures of two or more of these extremes (i.e., *mixed bonds*). Three mixed-bond types—covalent–ionic, covalent–metallic, and metallic–ionic—are also included on edges of this tetrahedron; we now discuss each of them.

For mixed covalent–ionic bonds, there is some ionic character to most covalent bonds and some covalent character to ionic ones. As such, there is a continuum between these two extreme bond types. In Figure 2.25*a*, this type of bond is represented between the ionic and covalent bonding vertices. The degree of either bond type depends on the relative positions of the constituent atoms in the periodic table (see Figure 2.8) or the difference in their electronegativities (see Figure 2.9). The wider the separation (both horizontally—relative to Group IVA—and vertically) from the lower left to the upper right corner (i.e., the greater the difference in electronegativity), the more ionic is the bond. Conversely, the closer the atoms are together (i.e., the smaller the difference in electronegativity), the greater is the degree of covalency. Percent ionic character (%IC) of a bond between elements A and B (A being the most electronegative) may be approximated by the expression

$$\%\text{IC} = \{1 - \exp[-(0.25)(X_A - X_B)^2]\} \times 100 \tag{2.16}$$

where X_A and X_B are the electronegativities for the respective elements.

Another type of mixed bond is found for some elements in Groups IIIA, IVA, and VA of the periodic table (viz., B, Si, Ge, As, Sb, Te, Po, and At). Interatomic bonds for these elements are mixtures of metallic and covalent, as noted on Figure 2.25*a*. These materials are called the *metalloids* or *semi-metals*, and their properties are intermediate between the metals and nonmetals. In addition, for Group IV elements, there is a gradual transition from covalent to metallic bonding as one moves vertically down this column—for example, bonding in carbon (diamond) is purely covalent, whereas for tin and lead, bonding is predominantly metallic.

Mixed metallic–ionic bonds are observed for compounds composed of two metals when there is a significant difference between their electronegativities. This means that

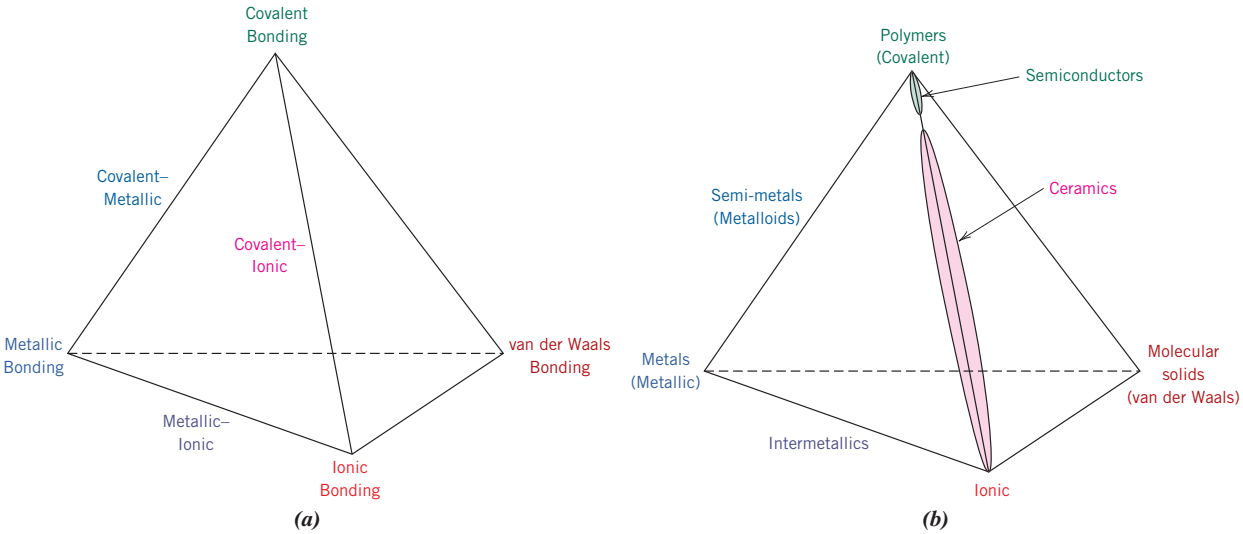


Figure 2.25 (*a*) Bonding tetrahedron: Each of the four extreme (or pure) bonding types is located at one corner of the tetrahedron; three mixed bonding types are included along tetrahedron edges. (*b*) Material-type tetrahedron: correlation of each material classification (metals, ceramics, polymers, etc.) with its type(s) of bonding.

some electron transfer is associated with the bond inasmuch as it has an ionic component. Furthermore, the larger this electronegativity difference, the greater the degree of ionicity. For example, there is little ionic character to the titanium–aluminum bond for the intermetallic compound $TiAl_3$ because electronegativities of both Al and Ti are the same (1.5; see Figure 2.9). However, a much greater degree of ionic character is present for $AuCu_3$; the electronegativity difference for copper and gold is 0.5.

EXAMPLE PROBLEM 2.3

Calculation of the Percent Ionic Character for the C-H Bond

Compute the percent ionic character (%IC) of the interatomic bond that forms between carbon and hydrogen.

Solution

The %IC of a bond between two atoms/ions, A and B (A being the more electronegative), is a function of their electronegativities X_A and X_B, according to Equation 2.16. The electronegativities for C and H (see Figure 2.9) are $X_C = 2.5$ and $X_H = 2.1$. Therefore, the %IC is

$$\begin{aligned}\%\text{IC} &= \{1 - \exp[-(0.25)(X_C - X_H)^2]\} \times 100 \\ &= \{1 - \exp[-(0.25)(2.5 - 2.1)^2]\} \times 100 \\ &= 3.9\%\end{aligned}$$

Thus the C—H atomic bond is primarily covalent (96.1%).

2.9 MOLECULES

Many common molecules are composed of groups of atoms bound together by strong covalent bonds, including elemental diatomic molecules (F_2, O_2, H_2, etc.), as well as a host of compounds (H_2O, CO_2, HNO_3, C_6H_6, CH_4, etc.). In the condensed liquid and solid states, bonds between molecules are weak secondary ones. Consequently, molecular materials have relatively low melting and boiling temperatures. Most materials that have small molecules composed of a few atoms are gases at ordinary, or ambient, temperatures and pressures. However, many modern polymers, being molecular materials composed of extremely large molecules, exist as solids; some of their properties are strongly dependent on the presence of van der Waals and hydrogen secondary bonds.

2.10 BONDING TYPE–MATERIAL CLASSIFICATION CORRELATIONS

In previous discussions of this chapter, some correlations have been drawn between bonding type and material classification—namely, ionic bonding (ceramics), covalent bonding (polymers), metallic bonding (metals), and van der Waals bonding (molecular solids). We summarized these correlations in the material-type tetrahedron shown in Figure 2.25*b*—the bonding tetrahedron of Figure 2.25*a*, on which is superimposed the bonding location/region typified by each of the four material classes.[11] Also included are those materials having mixed bonding: intermetallics and semi-metals. Mixed ionic–covalent bonding for ceramics is also noted. Furthermore, the predominant bonding type for semiconducting materials is covalent, with the possibility of an ionic contribution.

[11]Although most atoms in polymer molecules are covalently bonded, some van der Waals bonding is normally present. We chose not to include van der Waals bonds for polymers because they (van der Waals) are *inter*molecular (i.e., between molecules) as opposed to *intra*molecular (within molecules) and not the principal bonding type.